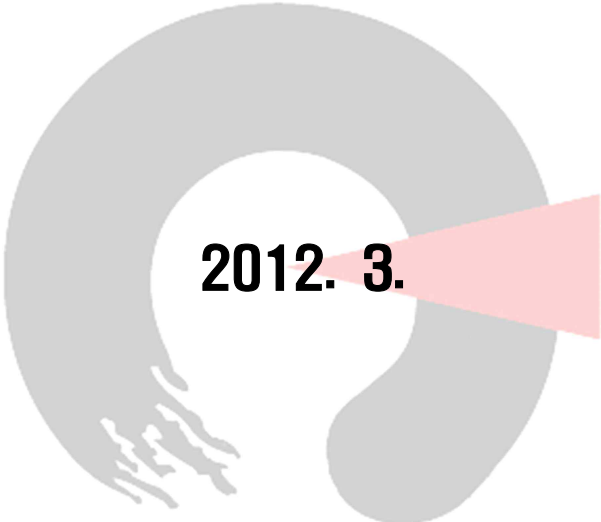


REQUIREMENT FOR PATENTABILITY



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KOREAN INTELLECTUAL PROPERTY OFFICE

Part III Requirement for Patentability

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Chapter 1 Industrial Applicability

1. Patent Act Article 29 paragraph (1)

Inventions that have industrial applicability are patentable unless they fall under either of the following sub-paragraphs:

- (1) inventions publicly known or worked within or outside of the Republic of Korea before the filing of the patent application; or
- (2) inventions described in a publication distributed in the Republic of Korea or in a foreign country before the filing of the patent application or inventions published through electric telecommunication lines as prescribed by Presidential Decree.

2. Purport

It is no doubt that all inventions should be industrially applicable since the purpose of the Patent Act is to contribute to the development of industry (Patent Act Article 1). In this regard, the Patent Act Article 29 paragraph (1) stipulates that an invention is patentable only if the invention is considered to industrially applicable. The term of "industry", the Patent Act Article 29 paragraph (1), shall be interpreted in a broad sense. In other words, the term industry is interpreted to cover all useful activities and practical technologies.

(Reference) Paris Convention Article 1(3)

Industrial property shall be understood in the broadest sense and shall apply not only to industry and commerce property, but likewise to agricultural and extractive industries and to all manufactured or natural products, for example, wines, grain, tobacco leaf, fruit, cattle, minerals, mineral waters, beer, flowers, and flour.

3. Relevant Provisions

Under Patent Act Article 29 paragraph (1), 「Invention which is industrially applicable」 is generally divided by the requirements of 「Invention」 and the

requirement of the invention 「that are industrially applicable (industrial applicability)」 . Therefore, under the examination guidelines, the requirements of Patent Act Article 29 paragraph (1) is defined by the requirement of 「Invention」 and of 「subjects that involve industrial applicability」 .

4. Statutory Inventions

Under the Patent Act Article 2 subparagraph (1), 「Invention means the highly advanced creation of a technical idea using the rules of nature」 , the invention shall satisfy the relevant provisions in order for a filed patent application to be statutory under the Patent Act.

However, 「a highly advanced creation」 is a relative concept to differentiate 「design device」 under the Utility Model Act from 「Invention」 under the Patent Act, and in practice, 「the highly advanced creation」 shall not be considered in the case of determining requirements of 「Invention」 .

4.1 List of Non-statutory Inventions

The decision of whether an invention falls under the Patent Act 2 subparagraph (1) is not without challenge, and the guidelines hereby exemplifies the types of non-statutory inventions in order to help determine whether the invention is patentable.

4.1.1 A law of nature as such

An invention is defined as a highly advanced creation of a technical idea using the rules of nature. So, a law of nature such as the second law of thermodynamics or the law of conservation of energy is not considered as a statutory invention.

4.1.2 Mere discoveries and not creations

A mere discovery is not deemed to be a creation because a discovery means to

find out laws which already previously existed in nature. One of the requirements for a statutory invention is to be a creation, and thus, mere discoveries, such as discoveries of natural things such as an ore or natural phenomena are not considered to be a statutory invention.

However, if things in nature, not mere discoveries, are isolated artificially from their surroundings, the methods, the isolated chemical substances or microorganisms are considered to be a statutory invention.

A use invention, which claims a new use of a known material in accordance with its inherent but newly found property, shall be dealt with distinctively from a "mere discovery of unknown property of a known material" in the Patent Act. That is to say, though a new use of a known material is non-statutory, if the new use are closely related, with non-obvious inventive activities, to the newly found property, the use invention may be patentable.

4.1.3 Those contrary to a law of nature

Those contrary to a law of nature (e.g.: perpetual motion) are not considered as a statutory invention because an invention must utilize a law of nature. If a matter necessary to define a claimed invention involves any means contrary to a law of nature, the claimed invention is not considered to be a statutory invention (Relevant court decision: Supreme court decision 1998.9.4 98Hu74 sentence).

4.1.4 Those in which a law of nature is not utilized

If a claimed invention uses any laws other than a law of nature (e.g. economic laws, mathematical methods, logics, cartography etc), arbitrary arrangements (e.g. a rule for playing a game as such) or mental activities (e.g. method for doing business as such, teaching skills as such, financial insurance scheme as such, tax code as such, etc.), the claimed invention is not considered to be statutory.

In the case where a claimed invention does not involve logics, mathematical principle as such or method directly using them but involves technical devices or a method which gives useful, concrete and tangible result by increasing or controlling the performance of a certain technical tools with the data, if the technical devices or technical methods are considered as universal, repetitive and objective, they are deemed as a statutory invention which uses technical idea utilizing a law of nature.

As stated above, the characteristics of the technology is to be taken into account as a whole in judging whether a claimed invention utilizes a law of nature. Therefore, even if a part of matters defining an invention stated in a claim utilizes a law of nature, when it is judged that the claimed invention considered as a whole does not utilize a law of nature, the claimed invention is deemed as not utilizing a law of nature. On the contrary, even if a part of matters defining an invention stated in a claim does not utilize a law of nature, when it is judged that the claimed invention as a whole is considered as utilizing a law of nature, the claimed invention is deemed as utilizing a law of nature.

(Example 1)

A method for designing cryptographs through the combination of Alphabets, numbers and signs.

(Example 2)

A method for creating a phonetic transcription of foreign languages comprising the step of: using the phenomena in which there is change in pronunciation formed by a set of vocal organs including the shape of throat and sound of tongue formed at pronouncing a certain word to indicate different pronunciation or characteristics of forming a phonetic transcription as the shape of lips changes.

(Example 3)

A method for comprehensive management for recycling garbages or waste comprising the steps of: distributing special bags bar-code stickers attached with

personal information of a person who disposes garbage or waste to citizens; inviting the citizens to separate general waste and place garbage in special disposal bag with the bar-code stickers attached; collecting the disposal bags on a daily basis and discharging them at a waste disposal site; and sorting waste or garbage at the waste disposal site, wherein in the case of wrongly sorted garbage, the citizen who disposed garbage in a wrong disposal bag are detected by the bar-code and warned not to do it again.

4.1.5 Skill

A personal skill which is acquired by personal practice cannot be shared with third parties as knowledge due to lack of objectivity, so it is not considered to be a statutory invention.

(Example 1)

A method of performing musical instruments, a method of throwing a spilt-fingered fast ball characterized in the way of holding the ball in fingers and throwing the same.

4.1.6 Mere presentation of information

A mere presentation of information where the technical feature resides solely in the content of the information and its main objective is to present the information is not considered as a statutory invention.

(Example 1)

An audio compact disc where the feature resides solely in the music recorded thereon, computer program listings, and image data taken with a digital camera, etc.

However, if the technical feature resides in the presentation of information, the presenting per se, the means for presentation and the method for presentation, might be considered as a statutory invention.

(Example 2)

A plastic card on which information is recorded with letters, numbers and signs embossed on it (a technical feature residing in the means for presentation)

4.1.7 Aesthetic creations

An aesthetic creation may contain a visionary feature as well as a technical feature. Therefore, its evaluation is subjectively made. An aesthetic creation itself (e.g. paintings and carvings as such) is not considered as a statutory invention. However, if the aesthetic creation is achieved by technical composition or other technical means, they are viewed as a statutory invention.

4.1.8 Computer programming language or computer program

A computer program is a mere list of orders to have a computer operated. Therefore, a computer program is not considered as a statutory invention. However, in the case of an invention where data processing with a computer program is specifically executed using a hardware, a data processing unit (machine) operating in association with the computer program, its operating method, and a computer readable medium carrying the computer program, the invention is viewed as a statutory invention.

4.1.9 Those whose outcome of the claimed subject matter is not achievable

An invention whose outcome of the claimed subject matter is not achievable and reproducible is not considered as statutory, even if the means to achieve the goal of the invention is sufficiently described. It does not mean that the possibility of reproduction of a filed invention should account for 100%. Even with less than 100% possibility, it is construed that the invention can be reproduced if it is certain that the outcome is achievable.

4.1.10 Incomplete Invention

A statutory invention shall be complete and a complete invention is defined as

an invention in which the subject matters shall be specified clearly and thoroughly objectively so that a person with ordinary skill in the art to which the invention pertains may easily replicate the invention to achieve the intended technical effect. The decision on whether an invention is complete shall be made by considering the invention as a whole such as its purpose, subject matters and operational effects of the invention indicated in the description of patent application in accordance with the state of the art at the time of filing.

If a subject matter lacks concrete means to solve the problem to be solved or if it is clearly impossible for the subject matter to solve the problem to be solved by any means presented in a claim, the claimed invention is not considered as statutory. However, in this case, the inventor can verify that the claimed invention solves the problem to be solved by means presented in a claim with an appropriate and concrete evidence such as reliable experimental data of third parties.

As one of the requirements to gain a patent right, an examiner should distinguish between the requirement of completing an invention and that of satisfying description requirement. An incomplete invention or an invention yet to be completed at the time of filing cannot be amended later to correct the defect after the application is filed. However, in the case of deficiency in the description, it is possible to correct the defect through the amendment because deficiency in the description applies to an invention which has improper description but is considered complete at the filing. Therefore, if it is unclear under which case the invention falls, it is desirable to preferentially notify the reason for refusal under the Patent Article 42 paragraph (3).

4.2 Notification of reason for refusal in the case of non-statutory invention

If an application falls under the scope of the non-statutory invention such as a law of nature, discovery, subject matters against the law of nature, subject matters not using the law of nature, mere presentation of information, aesthetic creations or incomplete invention, the reason for refusal shall be notified with the reason that the invention does not involve "inventions that have industrial applicability", citing the main paragraph of Article 29 paragraph (1).

4.3 Difference between statutory invention under Patent Act and device under Utility Model

Under the Patent Act Article 29 paragraph (1), the device (including a composite) and method can be a patentable subject matter. However, under the Utility Model Act Article 4 paragraph (1), a utility model may be granted only for devices that relate to the shape or construction of article or a combination of articles.

4.3.1 Article under the Utility Model Act

There is no definition prescribed about an article or articles described in Utility Model Act Article 4 paragraph (1). However, it is construed that a subject matter is generally considered as articles under Utility Model Act on condition that it is the object for trade having the shape in the space and the purpose of its use is clear.

The simple explanation about the shape or construction of an article or a combination of articles are as follows:

(1) Shape

"Shape" is external figuration expressed in the line, the surface, and so on. For example, the shape of the cam or the tooth shape of the gear etc.

(2) Construction

"Construction" is a construction constructed spatially and 3-dimensionally. It is expressed in the contour of articles. It is also shown in a ground plan and an elevation view, and in some cases a lateral view or a cross section diagram. A circuit of electronic products may be deemed to be an article under Utility Model Act.

(3) Combination

Two or more articles are spatially separated respectively which is not related whether being used or not, have independently fixed construction or shape, and moreover, show the use value for relating to each other functionally by using those, that is called "combination." For example, the fastening tools which consist of a bolt and a nut are a kind of combination.

4.3.2 devices that does not relate to the shape or construction of article or a combination of articles

A device of which the category is a process, a device of a composition, a device of chemical material, a thing which is not fixed in a certain shape, animal variety, plant variety do not fall under the scope of a statutory device under Utility Model Act.

(Reference)

If an independent claim is a device regarding the shape or construction of an article or a combination of articles and dependent claims define a material of the subject matter of the independent claim, We consider the dependent claim as the device relating to the shape or construction of an article or a combination of articles.

5. Industrially inapplicable invention

The following is a list of 「industrially inapplicable inventions」. However, upon noticing that a claimed invention does not meet the requirements for industrial applicability, the ground should be indicated as specifically as possible in the notification of reasons for refusal.

5.1 Medical practice

(1) List of industrially inapplicable inventions

① A method for treatment of the human body by surgery or therapy and a diagnostic method practiced on the human body are considered to be industrially inapplicable inventions. A surgical method on the human body by means of such a surgical device (e.g. scalpel) or a method for treatment of the human body with a medicinal substance, other than medical activities by medical doctors (including oriental doctors) or someone directed by medical doctors, is considered as medical activities.

② A method including medical activities even in one as the part of the whole step or as indispensable parts in claims is not considered as industrially applicable.

③ A method having surgical operations whose purpose is both therapeutic and non-therapeutic (e.g. cosmetic effects) or a method having surgical operations in which it is difficult to separate the therapeutic effect from non-therapeutic effect is considered as a method for treatment of the human body by surgery practiced on the human body. Therefore, such method is not considered as industrially applicable.

(2) List of Industrially Applicable Inventions

① A medical device to be used in the operation practiced on the human body or to be used in medical cure and diagnosis, and medical products as such are considered to be classified as industrially applicable.

② A method for operating a medical device or a measurement method with the medical device, when the medical device is newly invented, is considered as industrially applicable except when the method includes mutual reactions between a human body and a medical device and practically medical activities.

③ A method for treating samples that have been extracted from a human body (e.g., blood, urine, skin, hair, cells or tissue) or discharged from a human body (such as urine, excrement, placenta, hair and nail) and a method for gathering data by analyzing such samples are considered to be industrially applicable on

the assumption that they are composed of separate steps separable from medical practices.

(3) Matters to be attended to in examining inventions including medical practices

Even if it is possible for a method of treatment to be used for treatment, surgery or therapy on the human body, the method is considered to be industrially applicable if the method is limited to be practiced on the animal body excluded from being practiced on a human body.

5.2 Inventions that cannot be commercialized

An invention only for personal use, academic or experimental purposes are regarded as industrially inapplicable. On the other hand, despite inventions indicated above, an invention concerning marketable or tradable subject matter is considered industrially applicable.

5.3 Inventions that cannot be implemented

An invention which can not be implemented or practiced is not considered as an industrially applicable invention even if it works in theory.

(Example 1)

A method for preventing an increase in ultraviolet rays associated with the destruction of the ozone layer by covering the whole earth's surface with an ultraviolet ray-absorbing plastic film.

Even in the case where an invention has not been used at the time of the filing, the invention is considered as industrially applicable if it is possible to be used in the industry in the future. Under the principle of law that the invention should be industrially applicable, it is sufficient that the invention be industrially applicable in the future. The principle of law here does not mean that the invention would be deemed to be industrially applicable if it is possible to be used in the industry only because the relevant technology is gradually advanced.

Chapter 2 Novelty

1. Patent Act 29 paragraph (1)

Inventions that have industrial applicability are patentable unless they fall under either of the following sub-paragraphs:

- (1) inventions publicly known or worked within or outside of the Republic of Korea before the filing of the patent application; or
- (2) inventions described in a publication distributed in the Republic of Korea or in a foreign country before the filing of the patent application or inventions published through electric telecommunication lines as prescribed by Presidential Decree.

(Reference)

"inventions publicly known or worked in the republic of Korea" is revised into "inventions publicly known or worked within or outside of the Republic of Korea". The revision shows that the terms of being publicly known or worked is expanded to meet the global standard. The revised term is applied to applications filed on or after 2006.10.1.

2. Purport

The purport of the Patent System is to grant an exclusive right that is a reward for the disclosure of an invention. So, an invention already disclosed to the public shall not be given exclusive rights. Under the Patent Act Article 29 paragraph (1), prior to the filing of the patent application, (i) inventions publicly known, (ii) inventions worked (iii) inventions described in a publication, (iv) inventions published through electric telecommunication line as prescribed by President Decree do not involve novelty, and consequently they are not patentable.

3. Relevant Provisions

3.1 Publicly Known Invention

「A publicly known invention」 means an invention the contents of which have been known to an unspecified person without obligation of secrecy in the Republic of Korea or a foreign country prior to the filing of the application. The time of filing in the 「prior to the filing of the application」 refers to the exact point of time of filing, even to the hour and minute of the filing(if the invention is publicly known, the time is converted into Korean time). It does not mean the concept of the date of filing. 「unspecified persons」 refers to the general public who does not need to abide by secret observance duty.

(Example 1)

If an application is not laid open and is not published after registration, the invention of said application shall not be used as prior art under Article 29 paragraph (1) subparagraph (2) as the application is not a publication distributed in the Republic of Korea or a foreign country before the filing of the patent application. However, even if there is no publication of registration, the application shall be used as prior art under Article 20 paragraph (1) subparagraph (1) after decision of registration, if the application is made available for public inspection.

3.2 Publicly worked invention

「A publicly worked invention」 means an invention which has been worked under the conditions where the contents of the invention are to be publicly known or can potentially be publicly known in the Republic of Korea or a foreign country (Definition of "working" refers to the Patent Act Article 2). Also, 「being public」 means a situation where it is no longer kept in secret. So, even when a small fraction of inner part of an invention is kept in secret with regard to working of the invention, it shall not be considered as a publicly worked invention.

(Example 1)

Conditions where the contents of the invention are considered to be publicly worked include, for example, a situation where a person skilled in the art may

easily understand the contents of the invention by observing the manufacturing process associated with the invention at a plant that is exposed to an unspecified person. Conditions where the contents of the invention can potentially be considered to have been publicly worked include, for example, a situation where, although inner parts of the manufacturing facility cannot be known to an unspecified person (a visiting inspector) by merely observing its exterior view and the person cannot know the invention as a whole without knowing that inner parts, the person is allowed to observe the inner parts or can have the inner parts be explained to the person. (i.e. the request for observation or explanation is not to be refused by the plant.)

3.3 Invention Described in a Distributed Publication

3.3.1 Distributed publication

A publication is "a document, a drawing or other similar medium for the communication of information, duplicated by printing, mechanical or chemical methods, etc. for the purpose of disclosing the contents to the public through distribution". A "Distribution" in the context of the wording "disclosing the contents to the public through distribution" means placing a publication as defined above in the condition where unspecified persons can read or see it. It does not necessitate the fact of a certain person's actual access to such a publication.

Patent gazettes such as microfilm or CD-ROM should be considered as a distributed publication , since the public could refer to the contents of the film by using a display screen and obtain a copy of it.

Meanwhile, non patent documents which are stored in floppy discs, slides or presentations as well as microfilm or CD-ROM should be regarded as distributed publication, as far as they are produced to make available to the public.

3.3.2 Distribution

「A distribution」 in the context of the wording "inventions described in a distributed publication" means placing a publication as defined above in the

condition where unspecified persons can read or see it. It does not necessitate the fact of a certain person's actual access to such a publication.

3.3.3 Time of distribution

When the time of publication is indicated in a publication, it is presumed as follows:

① In the case the time of publication is indicated in a publication

(a) In the case where only the year of publication is indicated, the last day of that year;

(b) In the case where the month and year of publication is indicated, the last day of the month of the year; and

(c) In the case where the day, month and year of publication is indicated, that date.

② In the case where the date of publication is not indicated in a publication

(a) The distribution date of a foreign publication is presumed in light of the period normally required to reach Korea from the country of the publication, as far as the date of its receipt in Korea is clear.

(b) In the case where there is a derivative publication such as a book review, an extraction or a catalog, the date of distribution of the publication in question is presumed based on the publication date of the derivative publication.

(c) In the case where there is a second edition or a second print of the publication, the date of distribution is presumed to be the publication date of the first edition indicated therein, provided that the contents of the second edition accords with those of the publication.

(d) In the case where other appropriate information is available, the date of distribution is presumed or estimated there from.

(Example 1)

We already know that companies quickly access catalogues published by rival companies in order to acquire technical information of new products home and abroad, thanks to the advancement in transportation and thriving trade among

countries. It is socially accepted idea that catalogues are distributed as soon as they are published. Therefore, the claim that the published catalogues have not been distributed but kept in storage is not acceptable from our experience. In this regard, we make a decision that catalogues are distributed prior to the filing of the application as long as the evidence of bringing the cited reference into the country before the filing of the application is concrete (Supreme court decision 1992. 2. 14 1991Hu1410 sentence).

3.3.4 Invention described in a publication

「An invention described in a publication」 means an invention identified by the matters described or essentially described, though not literally, in a publication.

"Matters essentially described, though not literally, in a publication" means those directly derivable from the matters described, taking into consideration the common general knowledge.

(Example 1)

In order for a device to be described in a distributed publication, at least the structure of the device should be described. Therefore, if a device whose technical feature lies inside is merely exhibited in the form of photograph, it is not considered as a device described in a publication(Patent court 1992. 2. 14 1998Huh3767 sentence).

3.4 Inventions distributed to the public through electric telecommunication line

3.4.1 Purport

With the advancement of communication technologies such as the Internet, the number of technologies published over the internet has been dramatically increasing. It is suggested that we need to reflect the technological change on the patent system since technologies published on the internet can be considered as prior art in comparison with those released by the existing printed publication, with regard to public availability, propagation speed and the level of technology except for the possibilities that due to characteristics of internet, the date and the

contents of the publication may be altered after publishing.

Considering that 「Printed Publication」 under Article 29 paragraph (1) subparagraph (2) defined "copied documents, drawings and photographs which aims to be published through printing or the mechanical and chemical method" (Supreme court 1992.10.27 1998Hu3767 sentence), technologies published over the Internet has been considered as known technology defined in the Act Article 29 paragraph (1) subparagraph (1) not as the prior art stated in the printed publication defined in the Act Article paragraph (1) subparagraph (2).

However, according to the proclaimed revision on Article 29 paragraph (1) subparagraph (2) on February 3rd 2001, inventions available to the public over the internet whose date of publication and contents are entitled to public trust shall be given the same status as inventions published through printed publication.

In addition, a publication by electric telecommunication line as prescribed by President Decree are considered prior art under Article 29 paragraph (1) subparagraph (2) of the Patent Act. A publication by other electric telecommunication shall be entitled to be prior art as known art under Article 29 paragraph (1) subparagraph (1).

3.4.2 Requirement for Information available to the public through electric Telecommunication Lines under Article 29 paragraph (1) subparagraph (2) to be cited as prior art

(1) Inventions publically known to the public through Electric Telecommunication Lines

A telecommunication line includes public bulletin board, e-mail group using a telecommunication line as well as internet. Moreover, a new electric or electronic telecommunication method which would appear in the future as the technology advances, shall also be included.

Telecommunication lines do not always need to be physical lines. The term

telecommunication means transmission or reception of code, words, sound or image through wired, wireless, optic, or other electro-magnetic processes.

Information made available to the Public through a CD-ROM or a Diskette shall not be considered publication of technology through telecommunication lines but considered publication of technology through distributed publication.

(2) Invention Made Available to the Public

In order to cite an invention publically known to the public through electric telecommunication lines as prior art as in the case of the printed publications, the invention shall be "the one available to the public".

The "public" means an unspecified person who does not have to keep an invention secret and "available to the public" means situations where the invention can be seen by an unspecified person, and "available to the public" does not require actual access of the invention.

Even if an invention is published through electric telecommunication lines as prescribed by Presidential Decree, in a case where the invention is only accessible by a specific person and the invention is restricted to the public, the invention is not considered as being available to the public.

To determine whether information is an invention made available to the public, one needs to decide whether the information is made available on a Web site with a general search engine or whether the Web Site is encoded in such a way that it cannot generally be read. Only in the case where an information is considered as being available to the public, it can be cited as a prior art.

(3) Inventions published through electronic telecommunication lines owned by a person or an entity prescribed by the Presidential Decree

In order to cite technical information made available to the public through electric telecommunication lines as prior art as in the case of the printed publications, it is required that the cited technical information was published by "designated entities" under the Presidential Decree. Designated entities under the presidential decree shall fall under the following;

① Governments, local governments, foreign government, foreign local governments or international organizations.

Whether a certain entity falls under government or local government under enforcement decree of the patent act, relies on national government organization act or local government law.

Whether a certain entity falls under foreign government or foreign local government relies on its related Act and subordinate statute of each foreign country. For example, the electric telecommunication lines owned by Korean Intellectual Property Office (hereinafter refers to KIPO) especially, cyber bulletin is a typical electric communication line under Article 29 paragraph (1) subparagraph (2).

KIPO has granted the same status on inventions disclosed on Web-site owned by KIPO as the inventions opened in the publication to be cited as prior art, thus publishing inventions on the Internet as well as in a CD-ROM or written form in a faster and more economical way becomes possible. Under the former Patent Act, KIPO is required to publish all inventions only in a CD-ROM or written form to include those among prior arts. Under the current Patent Act, KIPO grants the same status of prior art on the inventions disclosed on the Internet as prior art in publications.

Also the term "International organization" is defined to include intergovernmental organizations but does not include nongovernmental organizations such as Asian Patent Attorneys Association. Intergovernmental Organization includes the United Nations, World Intellectual Property Organization(WIPO), World Trade Organization (WTO) and the European Union as well as regional patent offices such as European Patent Office, African Intellectual Property Organization, OAPI and African Regional Industrial Property Organization, ARIPO.

② National/Public schools under the Higher Education Act or foreign National/Public universities

National/public schools under Higher Education Act Article 3 refer to national schools established and run by government or public schools established and run by local self-governing groups among schools for providing higher education

under Higher Education Act Article 2 (universities and colleges such as industrial college, education college, specialized college, communication college, technological college and others)

Whether a certain college falls under "foreign national/public university" relies on its related Act and subordinate statute of each foreign country.

③ National/public research institutes in our country or foreign country

National/public research institutes in our country include research institutes including the inspection center and laboratory run by local self-governing group or government-sponsored research institutes.

Whether a certain institute falls under foreign national/public research institutes, relies on its related Act and subordinate statute of each foreign country.

④ Corporation designated and publically notified by the Commissioner of Korean Intellectual Property Office

Korea Invention Promotion Association(KIPA) and Organization for Data Management Center (Korea Institute of Patent Information) are designated as "Corporations established to conduct patent information and related works" of Enforcement Decree of the Patent Act Article 1bis subparagraph (4) under public notification about managing the electric telecommunication line corporations regarding patent information (Public notification of Korean Intellectual Property Office). Korea Invention Promotion Association(KIPA) and Korea Institute of Patent Information conduct delegated affairs offered by Korean Intellectual Property Office and both of the corporations are managed and governed by Korean Intellectual Property Office. Thus, the information from electric telecommunication line used in those corporations is reliable.

3.4.3 Finding the time of publication and the content of information

It is important to decide the time of publication and the content of information regarding published invention from the electric telecommunication lines. If an examiner finds the time of publication and the content of information regarding

said invention from the electric telecommunication lines under the Article 1bis Enforcement Decree of the Patent Act, he or she can cite the invention as prior art without any additional verification. Where an applicant is not convinced of examiner's recognition, the applicant shall argue against the indicated time of publication and the content of information.

If an examiner cites the invention available to the public from the electric telecommunication lines as prior art in the process of examination, he or she shall describe not only the source of the cited invention but the time of publication. If an examiner is not able to find the time of publication, he or she can cite the invention as prior art after convincing that the cited invention was published before the filling of the application (or priority date) through additional searches.

In the electronic telecommunication lines, the time of publication is defined as the time of posting the relevant technology on the electronic telecommunication lines. Therefore, in the case of citing the electronic technical information retrieved from the electronic telecommunication lines, the time of publication is the time the invention was posted on the electronic telecommunication lines, even in the case of previously published documentations.

3.4.4 Method of Citation

In the case of citing electronic technical information retrieved from the electronic telecommunication lines, the bibliographical items such as author, title, name of publication and pages (or drawings and graph) about the electronic technical information, as far as they have been known, shall be listed in the following order in compliance with WIPO Standards ST.14.

But, if a cited documentation is patented and the patented documentation is published through the Internet, an examiner is allowed to describe the cited documentation in the same way as patent official gazettes in the form of CD-ROM without having to describe the date of searching and web-site address.

3.4.5 Matters requiring attention in applying guidelines

(1) Treatment of other web-sites hyper-linked from web-sites under the

presidential decree

The electric telecommunication lines under Article 1bis Enforcement Decree of the Patent Act is deemed reliable. However, other web site hyper-linked through the electric telecommunication lines shall not be considered reliable under Article 1bis Enforcement Decree of the Patent Act. It is because its credibility regarding the time of publication cannot be guaranteed since the web-site is run by other entities.

(2) Difference between the electronic communication lines under the presidential decree and other electronic communication lines

A publically known invention under Article 29 paragraph (1) subparagraph (1) of the Patent Act is defined as an invention that is or could be publically known to an unspecified person. Therefore, inventions available to the public through the electronic communication lines other than those under the presidential decree, which is publically known to an unspecified person before filling an application, fall under 「the publically known invention」.

"inventions publically known through the electronic communication lines under the Article 1bis of the Enforcement Decree of the Patent Act" and other inventions have different status as prior arts even though the inventions are publically known through the electronic communication lines.

If an invention falls under the publically known invention under the Article 1bis of the Enforcement Decree of the Patent Act, an examiner can cite the invention as prior arts without additional searches to confirm the contents of technology and the date of posting. In the case of other publically known inventions, an examiner can cite the invention as prior art only when the examiner can confirm the date of posting on the electronic communication lines.

A publically known invention posted on web-sites not defined under Article 1bis of the Enforcement Decree of the Patent Act wherein the web-sites exclusively allow users in a certain area to connect (ex: web-sites exclusively allowing to connect only from computers in a certain US college) falls under publically known invention in foreign countries. Therefore, the invention can be cited as a prior art under the Article 29 paragraph (1) subparagraph (1) of the Patent Act.

However, the invention cannot be cited as a prior art under the Article 29 paragraph (1) subparagraph (2) of the Patent Act since the invention was not publically known through the electric communication lines under the Article 1bis of the Enforcement Decree of the Patent Act.

If an invention is publically known through the web-site (if the college above is national) under Article 1bis of the Enforcement Decree of the Patent Act, as the invention was not only published in foreign countries but also publically known through the electronic communication lines, the invention can be cited as a prior art under the Article 29 paragraph (1) subparagraph (1) and the Article 29 paragraph (1) subparagraph (2) as well.

4. Determining the Identity of Claimed Inventions

(1) The examiner shall determine whether or not a claimed invention is novel by judging whether the claimed invention falls under the inventions categorized in the provision of Article 29 paragraph (1) subparagraph (1) to (2). If a claimed invention falls under the inventions categorized in the provision of Article 29 paragraph (1) subparagraph (1) to (2), the invention is not novel. If a claimed invention does not fall under the inventions categorized in the provision of Article 29 paragraph (1) subparagraph (1) to (2), the invention is novel.

(2) The claims must describe the subject matter for which protection is sought. (Article 42 paragraph (4)) So, the decision over the identity of invention is determined by the identity of the matters described in the claims

(3) When there are two or more claims in an application, the determination over novelty should be made by each claim.

4.1 Defining invention disclosed in claims

4.1.1 General principle of defining inventions

(1) When the claim statements are clear, defining the claimed invention should be made as stated in the claim. The terminology described in the claims are

interpreted as having a general meaning and scope generally accepted in the technical field with the exception of the case wherein the terminology has a specific meaning which is explicitly defined in the description. The terminology should be interpreted in an objective and reasonable way by taking into consideration of its technical meaning, taken together with the common general knowledge at the time of filing, based on the general meaning of the terminology.

(2) In the case where the description of claims is clearly understood, an examiner should avoid limited interpretation just by referencing detailed description of the invention or drawings in finding technical features of invention.

In the case where the subject matters are not described in the claims but in the detailed description of invention or drawings, an examiner should interpret the invention as not being described in the claims. On the contrary, in the case where the subject matters are described in the claims, an examiner should consider the subject matters in claims when interpreting an invention.

It is possible to consider the detailed description of invention or drawings in understanding the subject matters disclosed in the claims but it is noted that an examiner should not examine the claims by applying subject matters not described in the claims. For example, where the subject matters described in the claims are more comprehensive than embodiments in the detailed description, novelty and inventive step should not be determined by interpreting the specific embodiments described in the detailed description as the claimed invention.

(Example 1)

In a case where 'cream' is described in the claims and 'the highly preserved cream which contains less moisture than bean-paste' is disclosed in the detailed description as an embodiment, as the term 'cream' generally refers to fat taken from milk, regardless of content of moisture, the claimed invention should not be interpreted to be limited to the embodiment of the detailed description since a person skilled in the art can clearly understand the term.

(Example 2)

In a case where 'thin film type probe' is described in the claims and 'a certain pattern is formed on the tip of the probe in the longitudinal direction' is disclosed in the detailed description, the claimed invention should not be interpreted to be limited to the certain pattern formed in the tip of the probe in the detailed description since the claimed invention is clearly defined as 'thin film type probe'.

(Example 3)

In a case where the rotation direction of brush roller is not disclosed in the claims but the subject matters of brush roller which rotates around a body of rotation is found in the drawings, the claimed invention should not be interpreted to be limited to the rotation direction of brush roller just by referring to the rotation direction described in the drawings.

(3) In a case where an applicant specifically defines a term in the detailed description to the extent that it is clearly understood that the term is different from any general meaning, in order to specify the term as a specific meaning not as general meaning in the technical field to which an invention pertains, the term is interpreted as a term with the specific meaning.

However, only the description of the specific concept of the term in the claims in the detailed description and drawings, does not fall under the specific definition aforementioned.

(Note)

A term in a patent specification is interpreted with the general meaning in the technical field and should be unified over the whole specification. However, if an applicant intends to use a certain term to have a specific meaning, an applicant is allowed to define the meaning of the term. So, the term can be simply interpreted according to the specific definition when the meaning of term is defined in the description(Supreme court 1998.12.22 97Hu990 Sentence).

(4) In a case where a term disclosed in the claims is obscure and unclear, an examiner should examine whether the claimed invention can be grasped in view of the detailed description, drawings and common general knowledge as of the

time of filing. If the invention can be grasped, the examiner can notify the applicant reasons of refusal on deficiency in describing specification and novelty collectively.

(5) If a claimed invention is not clear, even in view of the detailed description in the specification, the drawings and the common general knowledge as of the time of filing, examination of novelty is not conducted and the reason for refusal due to the deficiency of the detailed description is notified.

4.1.2 Principle of defining invention which includes special expression

(1) A product specified by its work, function, property, or characteristic (hereinafter referred to as the function, characteristic, etc.)

When describing claims, it is possible to state the structure, method, functions, materials or a combination of these factors for the purpose of clarifying which matters are subject to protection. When function, characteristic, etc. are disclosed in the claims to limit the subject matters of the claimed invention, an examiner should not exclude the function, characteristic, etc. from the features of the invention when interpreting the claims. When a claim includes an expression specifying a product by its function, characteristic, etc. such an expression should, in principle, be construed as every product that has such function, characteristic, etc., except when it should be construed otherwise because the expression is specifically defined in the detailed description. However, it is noted that there are also cases where a product described by its function, characteristic, etc. should not be construed as a specific product among all products that have such function, characteristic etc. by taking into account the common general technical knowledge at the time of the filing.

(Example 1)

In a case where 「means to selectively join plastic materials」 is disclosed, it is appropriate that 「the means to selectively join」 mentioned here should not apply to materials such as magnetics which is difficult to join with plastic material.

(2) The claim which includes an expression specifying a product by its use (limitation of use)

Where a claim includes an expression specifying a product by its use (i.e. limitation of use), the examiner should interpret the claimed invention only as a product specially suitable for the use disclosed in the claim, by taking into account the detailed descriptions in the specification and drawings and the common general technical knowledge at the time of the filing. Even if a product includes all technical characteristics described in the claims, an examiner should not regard the product as the product described in the claim when the product is not appropriate for the relevant use or when the product needs conversion to be used. For example, 「crane hook with a shape of ~」 merely indicates hook includes technical features with size and strength suitable for crane. So it is appropriate that the crane hook should be construed as a different product from 「fishing hooks」 with regard to the structure.

If a product with a limitation of use is not suitable for such use by taking into account the specification, drawings and the common general technical knowledge at the time of the filing, it is construed that a limitation of use has no impact in specifying an invention, thereby the limitation of use does not have influence in the determination of novelty.

(Example 1)

In the case where an embossing non woven fabric used in agriculture with limitations of weight and thickness is described in the claim and an embossing non woven fabric with the same numerical limitations is disclosed in a catalogue published prior to the time of filing, if it is considered that the claimed invention is not particularly suited to be used in agriculture, a limitation of use does not have influence in defining the claimed invention thus negating novelty based on the cited inventions in the catalogue.

(3) A product defined by its manufacturing process (product-by-process claim)

A product invention should be described in such a way that the technical

constitutions are directly stated in the claim, even if the manufacturing process of the product is disclosed in the product claim. Thus, an examiner should compare the claimed product itself defined by the description of the claim with a prior art published prior to the time of filing when determining novelty and inventive step, unless there is a special reason in the description of the claim. The special reason aforementioned should only be accepted by the Examiner in extremely exceptional cases such as when it is greatly difficult to define the product in the ordinary way in the relevant technical field.

Where a claim includes a statement defining a product by its manufacturing process, such a statement is construed as meaning a product per se unless it should be construed as different meaning according to the definition in the detailed description. If an identical product can be obtained by a different process from the one stated in the claim, the claimed invention is not novel where the product is publicly known prior to the time of filing. Thus, even if applicant's intention is to limit the claimed invention to only the product which is obtained by the particular process, such as a claim reading as "Z which is obtained solely by process A," the claimed invention should be treated in the same way aforementioned.

(Example 1)

In a case where 'panel formed by cutting process using an wave shaped blade of a knife' is described in the claims and the panel is the subject matter subject to protection, it is construed that there is no difficulty in directly specifying the structure of the panel in the technical field. It is appropriate to make a distinction between the panel defined by its manufacturing process and the claimed invention without taking into account of the manufacturing process. When comparing the claimed invention and the cited invention, both inventions show the same wave or cloud shape on the striped surface in the natural form. Therefore, the claimed invention is regarded as the same invention with the cited invention.

(4) Claims divided by the preamble and the body (Jepson type claim)

Even in the case of Jepson type claim which divides claims by the preamble

and the body, an invention should be specified as a whole including the preamble because the type of claims does not change the technical scope.

However, the subject matters described in the preamble are not considered to be publicly known just because of the fact that the subject matters are described in the preamble. The reason is that whether the subject matters in claims are publically known before the time of filing is a matter of facts and the claim type does not show that the matter of facts has occurred. Even if all subject matters described in the preamble are publically known, it is not appropriate to compare only the subject matters of the rest of the body with a prior art because the technical idea of the whole invention including the publically known subject matters of the preamble is the subject matter in determining patentability.

(Example 1)

In the case where the subject matters described in the body of the Jepson type claimed invention are anticipated in the cited invention but the subject matters of the preamble such as spark plug, ventilator are not described in the cited invention, it is not appropriate that said spark plug and ventilator are treated as publically known just because those are shown in the preamble. Moreover, an object of determining the patentability is the technical idea of the claimed invention as a whole body which includes the preamble. So, an examiner should not negate novelty based on the cited invention which does not include the technical features of the preamble.

(Note)

In the case of the description type which divides claims by the preamble and the body (the type called Jepson type claim), the preamble can be construed as various meanings such as ① limiting the technical field of inventions ② limiting the product applied by the technology of invention ③ excluding the scope of the right protected given that the invention is publically known. The body which is combined with the preamble is the technical characteristics of the claimed invention subject to protection.

4.2. Finding of a cited invention

Finding an invention cited in determining novelty (hereinafter referred to as "cited invention") under the Patent Act Article 29 paragraph (1) subparagraph (1), (2) are as follow.

4.2.1 Publically Known Invention

A "publicly known invention" means an invention the contents of which have been known to an unspecified person without obligation of secrecy in the Republic of Korea and a foreign country before the filing of an application. Finding a cited invention is basically carried out based on the matters publically known. Taking into consideration the common general knowledge as of the filing, if a person skilled in the art can easily arrive at the matters described in the invention, the matters are considered as being publically known.

(Reference)

The common general knowledge means technologies generally known to a person skilled in the art (ex, well known art or commonly used art). "Well-known art" means technologies generally known in the relevant technical field, e.g., those appeared in many prior art documents, those widely known throughout the industry, or those well-known to the extent needless to present examples. "Commonly used art" means well-known art which is used widely.

4.2.2 Publicly Worked Invention

A "publicly worked invention" means an invention which has been worked under the conditions where the contents of the invention are to be publicly known. Therefore, it is enough to decide whether the invention is publically worked without determining whether the invention is publically known.

"A publicly worked invention" means an invention which has been worked under the conditions where the invention is or can potentially be publicly known to an unspecified person through the medium of machinery or systems, etc. Therefore, the finding an invention can be carried out on the basis of the subject matters embodied in machinery or systems, etc. The matters directly derivable from the facts in view of the common general knowledge as of the working can also be

a basis for the finding of a publicly worked invention.

4.2.3 Invention Described in a Distributed Publication

"invention described in a distributed publication" means an invention which is or potentially be described in a publication. Being potentially described in a publication means that a person skilled in the art can easily recognize the invention. Such an invention can be considered as an invention described in a distributed publication.

4.2.4 General considerations in finding cited inventions

A manuscript for a journal of an academic society, in general, is usually kept secret against a third party, even after the receipt of the manuscript by the academic society. Therefore, the invention described in that manuscript is not considered a publicly known invention until its contents are released.

A company produces a catalogue to promote the company or to introduce and promote its products. Therefore, if the catalogue is produced, the catalogue is considered as a publication except when the catalogue is not published due to special circumstances.

In the case where the filing date of a patent application is the same as the date of the publication, the claimed invention does not lose novelty under the Article 29 paragraph (1) subparagraph (2) of the Patent Act , except when the filing time of application is clearly after the time of publication.

The time of publication of a thesis is the time when the thesis is distributed to an unspecified person in public or university libraries after the final thesis examination, except when the contents of the thesis are announced in an open space before the final thesis examination.

4.3 Method of Determining whether a Claimed Invention is Novel

The determination of novelty of a claimed invention is conducted by comparing the matters defining the claimed invention and the matters disclosed in the cited invention and extracting the difference between them. Where there is no difference between the matters defining a claimed invention and the matters disclosed in the cited invention, the claimed invention is not novel. Where there is a difference, the claimed invention is novel. The claimed invention is not novel when it is substantially or exactly identical to the cited invention.

「The substantially identical invention compared with prior arts」 means that there is no newly produced effect because the difference in the concrete means for solving problems is caused by mere addition, conversion or deletion of well-known or commonly used art and the difference between the claimed invention and the cited invention does not practically affect the technical idea of the claimed invention.

4.3.1 Determining novelty on invention with numerical limitation

An invention with a numerical limitation means that some part of the subject matters of an invention described in the claims is defined by specific numerical values. In the case where an invention in the claims includes a numerical limitation, a claimed invention is regarded as being novel when the claimed invention is not identical to the cited invention even when the numerical limitation is not considered.

When a claimed invention is identical to the cited invention except for numerical limitation, the determination of novelty comes under the following criteria.

(1) In a case where no numerical limitation is found in the cited invention while new numerical limitation is included in the invention described in the claims, the invention is regarded as novel. However, if the numerical limitation can be chosen by a person skilled in the art or it can be hinted in a cited invention in view of the common technical knowledge, novelty of the invention is denied in general.

(2) In a case where the numerical range of the invention described in the claims is included in the numerical range disclosed in a cited invention, it does not

negate novelty and the invention can be regarded novel by the criticality of the range of the numerical limitation. For the criticality of the range of the numerical limitation to be acknowledged, a remarkable change in the effect of the invention is required across the boundary of the numerical limitation and the following condition should be satisfied: 1) The technical meaning of the numerical limitation should be described in detail, 2) the embodiments in the detailed description or supplemental materials should prove that the range of the numerical limitation is critical. Generally, it should be objectively confirmed that the range is critical with experimental results which cover all range of the numerical limitation.

(3) In a case where the numerical range of invention described in the claims includes the numerical range of the cited inventions, novelty can be denied at once.

(4) In a case where the numerical range of the claimed invention is different from that of cited invention, novelty is regarded novel in general.

4.3.2 Determining novelty in parameter inventions

(1) A parameter invention is an invention in which an applicant creates a certain parameter which is not standard or commonly used for a characteristic value in physics or chemistry, parameterizes it arithmetically by using the correlation between plural parameters, and employs it as a part of essential element of the invention. Since the parameter invention may not precisely define the subject matter with the description of the claim itself, determination of the inventive step of the parameter invention should be performed only after figuring out the subject matter based on the detailed description, drawings and common knowledge.

(2) Novelty regarding a parameter invention is determined by interpreting the parameter itself as part of the claims, but it is important that novelty cannot be confirmed just because the parameter described in the claims is regarded as being novel. Novelty regarding an invention described in the claims is denied in

general if limiting the invention with the parameter just experimentally identifies nature or characteristics of a publically known product or there is a change just in expression by using parameter.

(3) In a parameter invention, if there is a 'reasonable doubt' that the claimed invention and the cited invention are identical, an examiner can wait written arguments or a certificate of experimental results after notifying the reason for refusal on novelty without comparing strictly the claimed invention with cited references because generally it is hard to compare the claimed invention with cited invention regarding determining novelty on parameter invention. If the reason for refusal is no longer kept by the applicant's arguments the reason for refusal is dissolved. But if the reasonable doubt is not dissolved, an examiner should decide the refusal of patent with regard to novelty.

(4) An examiner might have aforementioned reasonable doubt in the following cases: ① In a case when the parameter described in claims is converted with different definition and measurement method, and then the claimed invention is found to be identical with the cited invention. ② In a case when an examiner evaluates the parameter of a cited invention according to the measurement method in the description and obtains the same subject matter as that of claimed invention. ③ In a case when an embodiment in the detailed description of the claimed invention is identical to that of the cited invention.

(5) In a case when an examiner notifies the reason for refusal of a parameter invention, the examiner has to concretely describe the ground of reasonable doubt, and if necessary, the examiner can propose a way to overcome the grounds of rejection.

(6) The examination criteria described in (1)-(5) are not applied to a claimed invention when the parameter of the claimed invention is standard, commonly used or proved to be easily understandable by a person skilled in the art.

4.4 General considerations in determining novelty

(1) If an invention described in the claims and a cited invention are expressed in a generic concept or a specific concept, the following items should be considered in determining novelty:

① If an invention described in the claims is expressed in a generic concept and a cited invention is expressed in a specific concept, the invention in the claims is not novel. "Generic concepts" is defined as concepts integrating matters in the same family or the same genus, or a concept integrating a plurality of matters with the common characteristic.

(Example 1)

If a claimed invention is described as metal and a cited invention is described as Cu, the claimed invention is not novel.

② If a claimed invention is expressed in a specific concept and a cited invention is expressed in generic concept, the claimed invention has novelty. However, when an invention expressed in a specific manner can be directly derived from such a generic invention in consideration of the common general knowledge, the claimed invention's novelty is denied by defining an invention expressed in specific concept as a cited invention. An invention expressed in a specific concept cannot be derived from the inventions expressed in a generic concept, even if the invention expressed in a specific concept simply belongs to a generic concept or the elements of the specific concept can be presumable in the terms in generic concept.

(Example 1)

Silver is described in the claim as a superconducting cable material for electric power transmission and a cited documentation discloses a superconducting metal cable. If using silver as a cable material to activate super conductivity in the field of electric power transmission belongs to commonly known art, novelty of the claimed invention can be negated, as a person skilled in the art can conceive superconducting silver cable without undue difficulty.

(2) In determining novelty, the comparison shall not be conducted between a

claimed invention and a combination of two or more cited inventions. Determining patentability by a combination of two or more cited inventions is not related to novelty, but to inventive step. Except when a cited invention cites a distinct publication (ex: publication which provides detailed information of a technical feature, the distinct publication is regarded as a cited invention and able to be cited as determining novelty. When a dictionary and a reference are needed to interpret a term described in the cited reference, the dictionary and the reference are regarded as a cited reference and can be cited.

(3) If one or more inventions are described in a claim such as in a Markush type claim (multiple claims or features selectively cited or described, etc.), the reason of refusal can be notified on the ground that the invention does not involve novelty and an inventive step on each invention with one single prior art.

(4) In a case where there are more than two embodiments in a cited documentation, an examiner should not determine novelty by combining the two embodiments. Determining patentability through combination of cited embodiments is not a matter of novelty but inventive step. However, it is exceptional when one cited invention is obviously drawn from more than two embodiments in considering common general knowledge.

(5) In the case where the applicant admits the background technology to be publically known in a specification or written opinion, an examiner can determine novelty concerning a claimed invention by citing the background technology.

5. Inventions not considered to be publicly known

5.1 Patent Article 30

(1) Where a person who has a right to apply for a patent files a patent application for an invention within twelve months (six months in case of filing date being earlier than March 15, 2012) of date on which the invention falls

under any of the following sub-paragraphs, the invention is considered not to fall under any of the sub-paragraphs of Article 29 paragraph (1) when Article 29 paragraph (1) or (2) applies to the invention.

1. In the case where a person with the right to obtain a patent causes the invention to fall under either sub-paragraph of Article 29 paragraph (1) ; However, this provision does not apply where a patent application is laid open or a patent registration is published in the Republic of Korea or a foreign country under a treaty or law.

2. In the case where, against the intention of a person with the right to apply for a patent, the invention falls under either subparagraph of Article 29 paragraph (1).

(2) A person who intends to take advantage of paragraph (1) subparagraph (1) shall submit a written statement of that intention to the Commissioner of the Korean Intellectual Property Office when filing a patent application; the person shall also submit a document proving the relevant facts to the Commissioner of the Korean Intellectual Property Office, within thirty days of the filing date of the patent application.

5.2 Purport of the system

Under the Article 29 paragraph (1), a publically known invention before the filing of the application is not novel. However, under the Article 30, even if an invention is published before the invention is filed to obtain a patent, in the case where the necessary conditions is fulfilled, if the invention is not considered to be publically known, the invention cannot be cited as a prior art in determining novelty and inventive step under the Article 29 paragraph (1), (2). Under the regulation, the date of filing is not applied retroactively.

The purpose of regulation is to encourage an applicant to obtain a patent even after he or she publishes his or her own invention and to encourage early publication of an invention to help development of the national industry.

The regulation was amended on 3 March 2006. The amendment benefits an applicant when he or she publishes an invention both in the Republic of Korea

or a foreign country, except for the cases such as laying-open of application and publication of registration. The reason is that with the introduction of internationalism regarding publically known or used inventions, it is necessary to allow publically known or used inventions abroad to give benefits from the exceptional cases and that applicants tend to publish their thesis on the Internet not in the existing scientific journals. Moreover, as the application for the exceptional cases is limited to a certain type of being publically known, decision whether to apply put burdens on both applicants and examiners.

5.3 Requirements for inventions which are not considered to be publicly known

5.3.1 Where an invention is laid open by a person with the right to apply for a patent before the filing of the patent application

(1) Although an invention is laid open by a person with the right to apply for a patent before the filing of the patent application, the invention is not considered to be publicly known if the invention falls under either subparagraph of Article 29 paragraph (1) as prescribed in Article 30 paragraph (1) subparagraph (1) and meets the following requirements:

- (a) the invention is publicly known by a person with the right to apply for a patent
- (b) the invention is filed by a person with the right to apply for a patent within twelve months (six months in case of filing date being earlier than March 15, 2012) of the date on which the invention is publicly known (if the date on which the invention is publicly known is unspecified, the first day of the month and year of a disclosure may be applied.);
- (c) the purpose of being applied by the provision of Article 30 must be stated in the application; and
- (d) documents proving the relevant facts must be submitted within thirty days of the filing date of the application.

(2) To be considered as inventions which are not considered to be publicly known before the filing of the patent application, the inventions must meet either of the subparagraphs of Article 29 paragraph (1) along with above mentioned

requirements (a) to (d).

5.3.2 When an invention is publicly known, against the intention of a person with the right to apply for a patent

When an invention is publicly known against the intention of a person with the right to apply for a patent, it doesn't matter how the invention is publicly known. However, the person with the right to apply for a patent shall also file a patent application within twelve months (six months in case of filing date being earlier than March 15, 2012) of the date on which the invention is publicly known, without the need to state the purport of invention to take advantage of Article 30 in the application.

5.3.3 Distinction between the cases when an invention is publicly known by a person with the right to apply for a patent, and when the invention is publicly known against the intention of the person

(1) In filing a patent application claiming the exception of the claimed invention being considered to be publicly known, there may be two cases. The first case is when a person with the right to file for a patent causes the invention to be publicly known, and the second case is when the invention is publicly known against the intention of the person. Both cases have requirements in common that (i) the patent application be filed within twelve months (six months in case of filing date being earlier than March 15, 2012) of the date on which the invention is publicly known, and (ii) invention considered to be publicly known be examined on a claim by claim basis. However, there are differences in the person who publicly displayed the invention, the medium used for being publicly known and required documents proving the relevant facts.

(2) An invention being publicly known by a person with the right to file for a patent means an invention has been laid open by an inventor or the person's entitled successor. Notwithstanding the consent of the person with the right to apply for a patent, if the invention is laid open by a person who is not a successor to the right, the provision of Article 30 may not apply to the

invention. Meanwhile, the case of an invention being publicly known against the intention of a person with the right to apply for a patent is considered as being publicly known against the intention of an inventor or the person's entitled successor. Whether an invention is publicly known against the intention of a person with the right is determined in consideration of the right holder's clear intention to lay open the invention at the time when the invention is publicly known.

(3) Whether an invention is publicly known by a person with the right to apply for a patent or by a person against the intention of a person with the right to apply for a patent, the type of disclosure is not restricted. In the case of disclosure by a person with the right to obtain a patent, when the application has been laid open or registered in the Republic of Korea or in a foreign country according to the treaties and laws, the provision of Article 30 can not be applied.

(4) A person intending to take advantage of the provision of Article 30 for the reason that the invention is publicly known by the person with the right to apply for a patent shall state the intention in the application. If not, the person is not allowed to amend the application later to put the intention. Meanwhile, a person intending to take advantage of the provision of Article 30 for the reason that the invention is publicly known against his or her intention does not need to state the purport in the application.

5.4 The procedure to take advantage of the provision of Article 30

5.4.1 Where a person with the right to apply for a patent has laid open the invention before the filing of a patent application

(1) The patent application should be filed within twelve months (six months in case of filing date being earlier than March 15, 2012) of the date on which the provision of Article 30 paragraph (1) subparagraph (1) applies to the invention. In other words, the applicant shall state, when filing the application, that the person intends to take advantage from the provision of Article 30 paragraph (1)

subparagraph (1).

(2) The applicant shall submit documents proving the fact that the provision of Article 30 applies to the invention, within thirty days of the date of filing.

(3) For an international patent application, notwithstanding Article 30 paragraph (2), a person intending to take advantage of Article 30 shall state their intention in the application and submit the documents proving the intention within thirty days after the critical date (the date for submitting domestic documents) as prescribed in Article 201 paragraph (4) (Refer to Article 200 of the Patent Act and Article 111 of Special provision).

(4) It shall be proved that a person who disclosed an invention is the person who has the right to apply for a patent. Where the person who disclosed the invention is not the inventor or applicant, the applicant shall submit documents proving that, when displaying the invention, he or she is a successor to the person with the right to apply for a patent.

5.4.2 In the case that the invention is publicly known against the intention of a person with the right to apply for a patent

A person intending to take advantage of Article 30 shall prove that the invention is disclosed or used, against the intention of the person. A claim of the fact that the invention is publicly known and the requirements of proving the fact are individually judged on a case by case basis. Where an invention is publicly known against the intention of a person with the right to apply for a patent under Article 30 paragraph (1) subparagraph (2), it is not the case that the right holder voluntarily discloses the invention. So, there would be many cases that are difficult to prove how the invention was publicly known, as in the case of Article 30 paragraph (1) subparagraph (1). Therefore, the fact that the invention is publicly known against the intention of a person with the right to apply for a patent may be proved in various ways.

5.5 Examination to decide whether Article 30 applies to an invention

5.5.1 Formality examination

When a patent application claiming that the invention should not be considered to be publicly known is filed, the Korean Intellectual Property Office shall conduct a formality examination to identify whether the application is filed within twelve months (six months in case of filing date being earlier than March 15, 2012) of the date on which the invention is publicly known and whether the invention is publically known by a person with the right to apply for a patent. Whether the invention publicly known is identical to the invention filed or not is examined at the substantial examination procedure.

Even if a person submits a document proving that an invention is displayed at an exhibition, if it is difficult to identify the invention displayed at the mentioned exhibition, the Korean Intellectual Property Office may invite the applicant to submit a complementary document to identify the fact. However, if the organizer of the exhibition, title, date, and venue of the exhibition are in doubt, the Office may ask the applicant to add documents to prove the fact.

Notwithstanding the submission of the written intention to take advantage from the provision of Article 30, if the claim falls under any of the following subparagraphs, the office gives the applicant a notice to correct defects.

- (a) when a person who disclosed an invention is different from the person who filed or invented the invention;
- (b) when the patent application is filed more than twelve months (six months in case of filing date being earlier than March 15, 2012) after the disclosure date of laying open the invention;
- (c) when an applicant puts the wrong disclosure date and type of the disclosure on the application and puts incorrect information on the documents proving that the invention is not considered to be publicly known; or
- (d) when the submitted documents are not enough to prove the relevant fact

Notwithstanding the notification, when the applicant does not make an amendment within a designated period, the office invalidates the claiming

procedure under Article 30(in this case, the filing procedure is effective).

5.5.2 Substantial examination

An examiner reviews all of the submitted documents and examines the invention if defects are not found in the application. The disclosures that fall under any of the provisions of Article 30 are recognized as non-prior art in the examination, where Article 29 paragraph (1) or (2) applies to the invention claimed in the patent application. However, when the procedure of claiming an invention to be not to be considered to be publicly known fails because defects are found in the application and documents, the submitted documents may be used as prior art.

5.5.3 Matters to be attended to apply Article 30 to the invention

(1) When a person with the right to apply for a patent discloses an invention several times before the filing of a patent application, the provision of Article 30 may apply to the all acting of disclosure if the acting is legitimately applied to the procedure of Article 30 paragraph (1) subparagraph (1). If an acting of disclosure is inseparably related to one or more disclosures, the applicant may be exempted from the submission of a document proving the relevant facts, in the second disclosure and after the second disclosure. In this case, the period of twelve months (six months in case of filing date being earlier than March 15, 2012) prescribed in Article 30 will be calculated from the earliest date of publication. The disclosure which is inseparably related to one more disclosures is prescribed in the Article 30 paragraph (1) subparagraph (1) and the applicable cases are as follows:

(a) an examination which takes two or more days; (b) examination and explanation distributed on the date of examination; (c) the first edition and second edition of a publication; (d) the collections and oral presentation of the collections in a society; (e) oral presentation and a lecture booklet; (f) a lecture tour; and (g) display in an exhibition and catalog of the displayed product.

(2) When there are an invention (A) which is filed for claiming that it is not considered to be publicly known and another invention (B) which is the same as

invention but laid open by a third person between the date of disclosing the invention (A) under Article 30 paragraph (1) subparagraph (1) and the filing date of the invention (A), an examiner shall refuse the application of invention (A) for the reason of lacking novelty, except the obvious fact: the laying open of the invention (B) was made by learning from the disclosure (A).

The laying open of invention which was made by a third person after learning from a disclosure, which is not considered to be publicly known, covers the case when a third person reproduces the invention which was publicly known by a person with the right to apply for a patent at an examination, publication, announcement in an academy, and display in an exhibition. After sending an applicant a notice of reasons for refusal for the above reason, the examiner shall decide to refuse an invention if the applicant can't establish the fact that the a third person laid open the invention after learning from a disclosure which is not considered to be publicly known, or that the invention is filed against the intention of the applicant.

(3) When a person filed a patent application A within twelve months (six months in case of filing date being earlier than March 15, 2012) of the date on which the invention is publicly known and took advantage from the provision of Article 30, and on the same filing date, a third person filed a patent application B with the same invention as A, Article 36 paragraph (2) applies to applications A and B. It means that A and B are related to the same invention which are filed on the same date, and the applicants of A and B should reach an agreement on who will obtain a patent for the invention. Moreover, in consideration that B is filed after the invention is publically known, B is deemed to lack novelty and therefore the applicant who filed B may not obtain a patent, without applying the provision of Article 36. In this case, the examiner shall instruct both applicants to report on the results of the consultation under Article 36 paragraph (6), not to give a notice of refusal and decide the refusal because B lacks novelty. According to the provision of Article 36, the examiner shall let both applicants know that only the person agreed upon by all the applicants after consultation may obtain a patent for the invention. Upon the examiner's instruction for consultation, where the applicant who filed the application B withdraws the filing, the applicant who filed the application A

may obtain a patent.

(4) When a patent application is filed by a person with the right to obtain a patent and the application is published in the patent gazette, Article 30 of the Patent Act does not apply to the invention.

Under the normal procedure of filing a patent, the Commissioner of the Korean Intellectual Property Office shall lay open a patent application in the patent gazette after the prescribed date of filing under Article 64 paragraph (1). Where an application is laid open according to the procedure of filing, the laying open is not the voluntary intention of the applicant. In this case, therefore, the provision of Article 30 is not applied. However, before an invention is laid open, the invention may be withdrawn or revoked or finally rejected. After the procedure of filing has been completed, if the Office mistakenly lays open an invention, the disclosure of the invention is regarded as the one against the intention of the applicant. In this case, the applicant may take advantage from the provision of Article 30.

(5) When an applicant files a patent application claiming priority under the relevant treaty, to take advantage of the provision of Article 30, the applicant shall file the patent application in the Republic of Korea, within twelve months (six months in case of filing date being earlier than March 15, 2012) of the date of acting to which Article 30 apply. However, with regard to a patent application which contains a priority claim based on a patent application filed in the Korean Intellectual Property Office, the applicant may take advantage of the provision of Article 30 if the earlier application is filed within twelve months (six months in case of filing date being earlier than March 15, 2012) of the date of publication, even though the subsequent application is not filed within the twelve months (six months in case of filing date being earlier than March 15, 2012).

(6) The provision of Article 30 paragraph (1) subparagraph (1) is also applied to the following laying-open actions: (a) when a person with the right to obtain a patent asks to disclose an invention to a third person and the invention is disclosed by the third person; and (b) when the third person discloses an

invention by referring after obtaining permission from a person with the right to obtain a patent (implied consent is also approved).

Above mentioned case (a) covers the following cases: (i) when a person with the right to apply for a patent commits the disclosure of an invention to a third person, including the case when the name of inventor or the right holder is stated, and (ii) when a person with the right to apply for a patent sends the press release or script of an invention to a newspaper and the invention is laid open; in this case, although the name of inventor or the right holder is not stated, the fact that the person who wrote to the newspaper is the right holder should be proved.

Moreover, a disclosure when a person refers to an invention of a person with the right to obtain a patent covers the following cases: (i) when a person, who has not the right to obtain a patent, clarifies the inventor or company for which the inventor works in paper or article and refers to the invention, and (ii) when a company for which the inventor works discloses the invention in printed manner such as catalog.

Meanwhile, when two or more persons jointly make an invention or file an application, among presenters who disclosed the invention, at least one person is the same with the inventor or applicant, the provision of Article 30 may apply to the invention without any need to prove that the inventor or applicant is the person who disclosed the invention. However, if there is no relation between the inventor or applicant and the person who disclosed the invention, the applicant is required to submit the following documents: (i) a document proving the fact that a person who presents an invention is the person with the right to obtain a patent (ii) a document proving that an invention is disclosed by a request from the person with the right to obtain a patent and (iii) a document proving that a person who presents an invention got permission for referring to the invention from the person with the right to obtain a patent.

Chapter 3 Inventive Step

1. Patent Act, Article 29, paragraph (2)

Notwithstanding paragraph (1), if an invention could have been easily made, before the filing of a patent application, by a person skilled in the art to which the invention pertains based on an invention or inventions referred to in each subparagraph of paragraph (1), a patent for such an invention may not be granted.

2. Purport

The purport of Article 29, paragraph (2) of the Patent Act is not to grant a patent to inventions that could have been easily made by a person skilled in the art, since granting a patent to such inventions does not contribute to and even hamper the technological progress. In other words, granting a patent to slightly advanced arts leads to literally giving exclusive rights to the same prior arts. This runs counter to the purport of the Patent Act, which gives an inventor exclusive rights in exchange for the publication of a new technology, and by doing so limits the possibilities of a third party of accessing the technology.

(Reference)

The term "inventive step" is not defined in the Patent Act. However, if an invention could have been easily made, the invention does not have an inventive step. Otherwise, the invention has an inventive step under Article 29, paragraph (2) of the Patent Act.

3. Relevant Provisions

3.1 Prior to filing of a patent application

「Prior to filing of a patent application」 does not refer to the simple concept of the filing date, but means the definite time, in hours and minutes at the time of filing. For instance, if an invention is publically known outside Korea and the

point of time when the invention has become publically known precedes the filing time of a patent application in Korean local time, then, this invention is considered as a cited invention under Article 29, paragraph (1) of the Patent Act.

3.2 A person skilled in the art

"A person skilled in the art to which the invention pertains" (referred to as "a person skilled in the art" hereinafter) refers to a hypothetical person who has common general knowledge in the art to which the claimed invention pertains and the ability to use ordinary technical means for research and development (including experiment, analysis, and manufacture); who has the ability to exercise ordinary creativity in selecting materials and changing designs; and who is able to comprehend based on his/her own knowledge all technical matters regarding the state of the art in the field to which a claimed invention pertains at the time of filing a patent application. In addition, an expert in the technical field is one able to comprehend based on his/her own knowledge all technical matters in the technological field relevant to a problem to be solved by the claimed invention.

"The state of the art" at the time of filing of a patent application includes "an invention(s) referred to in any of the subparagraphs of Paragraph (1)," the common general knowledge, and other publicly known technical matters. It also relates to all types of information relevant to the technical field of the invention described in the claims, including ordinary methods to conduct daily works and experiments.

3.3 Invention that could have been easily made

「Where an invention could have been easily made by a person skilled in the art based on an invention(s) referred to in each subparagraph of paragraph (1)」 refers to whether a person skilled in the art could have easily conceived the invention described in the claims by exercising ordinary creativity or based on motivation induced from the invention(s) that is(are) publically known prior to the filing of the patent application.

4. General principles of determining the inventive step

(1) When determining the inventive step, it is decided whether 「an invention described in the claims」 as filed could have been easily made by a person skilled in the art based on an invention(s) defined in Article 29 paragraph (1) of the Patent Act (hereinafter refer to as the 「cited invention(s)」), prior to the filing of the patent application. If 「the invention in the claims」 could have been easily made by a person skilled in the art, alone or by combining the cited inventions, the invention in the claims is not considered to has an inventive step.

(2) When there are two or more claims in an application, the determination should be made for each claim.

(3) Notifying applicants of the reasons for refusal regarding novelty is different from notifying applicants of the reasons for refusal regarding the inventive step. However, when it is determined that an invention is not novel, it is allowed to notify applicants of the reason for refusal along with the reasons for refusal regarding the inventive step.

(Reference)

The inventive step of the patent application is based on whether the claimed invention is novel. Therefore, determination whether the claimed invention is novel compared to published inventions should be distinct from determination whether the claimed invention could have been easily made by a person skilled in the art. Therefore, in order to determine the inventive step of the claimed invention, determination of novelty should come first (Supreme Court's decision 1992. 6. 2. 91Ma540).

(4) Regarding a claim which contains more than two inventions including a Markush group type (including the case when multiple claims or elements are selectively recited), if the examiner notifies the applicant of the reasons for refusal with regard to a certain invention, the examiner needs to precisely point out the invention along with the reasons for refusal with regard to novelty and

the inventive step.

5. Method of determining the inventive step

The examiner shall make efforts to consider the overall state of the art that a person skilled in the art would consider at the time of filing an application and, at the same time, shall thoroughly consider the purpose, technical structure, and advantageous effects of the invention while paying attention to the opinion of the applicant, comprehensively determining whether the claimed invention has an inventive step in consideration of its specific purpose and effectiveness, and focusing on the difficulty of the technical structure of the claimed invention.

Determination of the inventive step shall be done in consideration of ③ whether, from the point of view of one or ordinary skill the art, the claimed invention has any advantageous effects over a cited prior art while mainly focusing on ① whether the cited prior art provide any motivation to a person skilled in the art to arrive at the subject matter of the claimed invention or ② whether the difference between the subject matter of the prior art and that of the claimed invention can be considered as a mere exercise of ordinary creativity.

5.1 Procedure of determining the inventive step

The procedure of determining the inventive step is as follows.

(1) The examiner specifies an invention for which an applicant requests grant of a patent. The method of specifying the claimed invention is the same with that of 「determining novelty」 in Chapter 2.

(2) The examiner specifies cited inventions. The method of specifying the cited inventions is the same with that of 「determining novelty」 in Chapter 2. The examiner shall specify the cited inventions from the point of view of a person skilled in the art, on the assumption of the common technical field and technical problems of the claims of present invention.

(3) The examiner chooses a cited invention which is the closest to the claimed invention and makes a clear difference by comparing the cited invention with the claimed invention. In doing so, the examiner shall take into consideration the structural combination of the elements of an invention. More specifically, structurally combined elements of an invention shall be compared as one integrated unit (without being separated) with their corresponding elements in the cited invention.

(4) The examiner determines whether an invention described in the claims could have been easily made by a person skilled in the art, in view of cited inventions or the common general knowledge before the filing, even though there is a difference between the claimed invention and the cited inventions

5.2. Selection of the cited invention

(1) A cited invention, which is the object of comparison with a claimed invention in the determining the inventive step, shall be, in principle, selected from the same technical field as or from a reasonably relevant technical field to the problem, effect, and use of the claimed invention. The same technical field shall refer to, in principle, the industrial field where the invention is applied, but shall also refer to the technical field that can be inferred from the effects or functions of some (or all) comprising elements of the invention. Even if the prior art is in a different technical field from the invention described in the claims, the prior art can be recognized as a cited invention if the prior art might be applied to other technical fields or used by the applicant in the process of solving a specific technical problem.

When a claimed invention is compared to the prior art in a different technical field, the examiner should take into account when citing documents the relevance of two technical fields, the close similarity of a problem to be solved, and the close similarity of a function, work or operation.

(Example 1)

The claimed invention relates to a container cap, which can seal or unseal a

container for plant nutrition easily and completely. The cited invention discloses an eruption closure assembly to be used for liquids of different viscosities. The technical field of the claimed invention is similar to that of the cited invention since both inventions relate to an apparatus for sealing or unsealing a liquid container. Hence, the eruption closure assembly is properly chosen as a cited invention.

(Example 2)

The technical fields of an umbrella and a parasol are not exactly the same, but are considered to be proximate to each other since the upper cover can be unfolded with the supporting pole at its center. Hence, the inventive step of the umbrella can be denied by citing the parasol as a prior art.

(Example 3)

The claimed invention relates to a method of preventing damages due to harmful insects by trunk injection, which is a method of injecting medication into a tree and filling a bore after the injection. Cited invention 1 relates to a method of disposing of a bore in a tree after trunk injection. Cited invention 2 relates to a method of injecting antibiotic into trees to eradicate insects through trunk injection. As there are enough grounds to consider that cited inventions 1 and 2 and the claimed invention are in the same technical field, the inventive step of claimed invention can be denied due to cited inventions 1 and 2.

(2) 「The closest cited invention」 means the most relevant cited invention in relation to a claimed invention among several selected cited inventions that a person skilled in the art would choose and which discloses most of the technical features of a claimed invention. Hence, it is desirable to choose the closest cited invention from among inventions that are in proximate technical field or have the same effect, use, or relate to a technical problem to be solved that is identical or similar to that of the claimed invention.

(Example 1)

An umbrella can be the closest cited invention in denying the inventive step of a parasol because improving the portability of a foldable object by making its

size smaller when folded is a common technical problem to be solved in the relevant technical field.

(3) When there is a description in a cited reference whereby the technical idea of the claimed invention cannot be easily conceived, the cited reference is not eligible as a cited invention. However, regardless of the description in the cited reference, which is a *prima facie* case to determine that the technical idea of the claimed invention cannot be easily conceived, the eligibility of a cited invention shall be maintained if it is possible to arrive at the technical idea of the claimed invention from other aspects such as a close relation between technical fields, a close similarity of a function, work, or operation, etc.

(Example 1)

The claimed invention relates to a probe card and is in the same technical field with cited invention 1 which comprises all technical features of the claimed invention except that the cited invention does not disclose a means to control the overall flatness. However, there is no description in the cited invention 1 to preclude the adoption of such a means to control the overall flatness, nor it is technically difficult to introduce such a means considering its technical structure. Hence, the inventive step of the claimed invention can be denied by combining the technical features of cited invention 1 and a means of controlling the orientation of a substrate disclosed in cited invention 2 (which is in the same technical field as the claimed invention).

(4) If the applicant admits in the specification that a technology presented as prior art is publicly known prior to the filing of the application, the technology may be properly selected as a cited invention at the time of filing, in determining the inventive step of a claimed invention.

(5) Even though the prior art constitutes an incomplete invention, it can be cited in determining the inventive step

(Example 1)

The claimed invention relates to a pharmaceutical compound to treat

neuro-degenerative disorders by using an estrogen compound alone. A person skilled in the art can easily recognize from the cited invention that sexual hormones such as estrogen are effective for curing neuro-derogative disorders. And if this fact is not contrary to the technical common sense at the time of filing the application, the cited invention can be used as a prior art to determine the inventive step even if some defects exist in the description of the cited invention due to insufficiently disclosed pharmaceutical effects and real experiments.

6. Concrete method of determining the inventive step

6.1 Probable cause or motivation

The following cases can be strong grounds for determining that a person skilled in the art would have been led to the claimed invention based on the cited invention; suggestions shown in the disclosures of the cited inventions, close similarity of a problem to be solved, close similarity of a function, work, or operation, close relevance of technical fields.

6.1.1 Suggestions shown in the disclosures of the cited inventions

Suggestions shown in the disclosures of the cited inventions relevant to a claimed invention can be strong grounds for determining that a person skilled in the art would have been led to the claimed invention.

(Example 1)

The claimed invention discloses a technical method of establishing a condenser, a motor, and a compressor in an airtight cooling apparatus. The cited invention discloses a method of setting up the relevant structure of a cooling compressor in an airtight cooling apparatus. The cited invention differs from the claimed invention only in that the cited invention does not specifically mention a heat exchanger built in the airtight cooling apparatus of the claimed invention. If the cited invention implicitly suggests the heat exchanger built in the airtight cooling apparatus, which is a relevant component of the cooling apparatus, the technical

feature of claimed invention is merely a matter of design option when the general technical knowledge in the relevant field of the art is applied.

6.1.2 Close similarity of a problem to be solved

(1) A close similarity of a problem to be solved can be a strong ground for determining that a person skilled in the art would have been led to the claimed invention by applying or combining cited inventions.

If the technical problems to be solved described in the claimed invention and cited invention are not in the same technical field, the examiner decides whether the technical problem of the claimed invention is obvious in the relevant field of the art or easily deducible in light of technical common sense and whether that reasoning can be used as a ground for denying the inventive step by scrutinizing the technical problem.

(Example 1)

The claimed invention discloses an animal-shaped winter cap which creates a unique fashion style and offers protection against cold by covering not only one's head and ears but also the areas around the neck, cheeks, and lips. Cited invention 1 relates to an animal-shaped winter cap, and cited invention 2 is directed to a mask hood to protect one's face by covering all parts of 'the face except for the eyes in the event of extremely cold weather. In this case, the technical fields of the cited inventions are identical or proximate to that of the claimed invention. Moreover, problems to be solved by the claimed invention and its solutions seem to be suggested in each of the cited inventions. Hence, there seem to be no difficulty in combining the cited inventions, and therefore the claimed invention would have been readily derived from the cited inventions by a person skilled in the art.

(Example 2)

The claimed invention is directed to a snap action diaphragm to adjust snap-action of a diaphragm by controlling the degree of a slope of an outer circumference of the diaphragm by applying power on it. The cited invention

relates to a thermo-start which is activated in accordance with a temperature change. The technical problems of both claimed and cited inventions correspond to each other in that both inventions disclose a method to control the snap-action of a diaphragm. However, the two inventions differ from each other in that the diaphragm of the claimed invention is activated according to pressure changes, while that of the cited invention is set in motion in accordance with temperature changes. Nonetheless, the inventive step of the claimed invention would be denied if the difference does not have any significant influence over the gist of claimed invention and a person skilled in the art can easily apply a thermally actuated method to the pressure actuated diaphragm without exercising any creative thinking.

(2) Even in the case of a cited invention with a different problem compared to a claimed invention, if it is obvious that a person skilled in the art can easily arrive at a the claimed invention through a mere exercise of ordinary creativity, the inventive step of the claimed invention can be denied.

(Example 1)

The claimed invention relates to a carbon disc brake with grooves designed to prevent the attachment of water drops on its surface. Cited invention 1 discloses a carbon disc brake, and cited invention 2 shows a metal disk brake with grooves designed to remove dusts from its surface. These technical problems are not exactly the same, but a person skilled in the art would readily arrive at the carbon disk brake with grooves by simply combining the technical feature of cited invention 2 with the carbon disk brake of cited invention 1 without exercising any creative thinking, thereby the inventive step of the claimed invention can be denied.

6.1.3 Close similarity of a function, work, or operation

If a close similarity in a function, work, or operation exists between a claimed invention and a cited invention or between cited inventions, there can be a well-founded ground that a person skilled in the art would have arrived at the claimed invention.

(Example 1)

The claimed invention discloses a filtering apparatus for home use, with a specially structured filtering part. The cited invention relates to a home filtering apparatus, which is exactly the same as in the disclosure of the claimed invention except for the structure of the filtering part. Cited invention 2 is directed to a filtering apparatus for an automobile, with the same structured filtering part as that of the claimed invention. The filtering apparatuses described in cited inventions 1 and 2 are identical to that of the claimed invention with respect to their functions and operations. Considering that the claimed invention is not in a different technical field from the cited inventions in terms of the generally required technical problems, applying the filtering part described in cited invention 2 to the filtering apparatus in cited invention 1 is deemed to be obvious to a person skilled in the art.

6.1.4 Close relation of technical fields

A publically known technical means for solving the technical problem of the claimed invention in a related technical field can be a strong ground for considering that a person skilled in the art could have achieved the claimed invention.

(Example 1)

If a cited invention discloses gloves with a similar structure to claimed socks, as gloves and socks belong to similar technical fields and they are related to each other, a person skilled in the art could easily apply the composition of socks to gloves.

6.2 Mere exercise of ordinary creativity of a person skilled in the art

A common improvement based on general applications of a known art, reasoning based on known physical properties, or referring to other technical field to solve a known problem fall into the category of ordinary creativity of a person skilled in the art. Among exercises of ordinary creativity of a person skilled in the art

are selecting an optimal material from publicly known materials in order to achieve a specific object, optimizing a numerical value range, replacing with equivalents, a workshop modification of a design in applying a specific technology, partially removing technical features and simply changing the use. When the differences between the claimed invention and the cited invention under comparison falls only under these categories, it is usually considered that a person skilled in the art could easily arrived at the claimed invention, unless there is another ground for assessing the inventive step.

6.2.1 Replacement with Equivalents

An invention consisting of a substitution in a known device of a feature with a known feature which has the same function as the substituted feature and is an equivalent thereof is not considered as having an inventive step if it fails to exhibit an unexpected advantage.

In order to acknowledge the replacement with equivalents as a mere exercise of ordinary creativity of a person skilled in the art, it should be justified that the replacement is obvious to a person skilled in the art at the filing of the application in addition to the fact that the substituted known feature functions as an equivalent. The examiner is allowed to submit evidence that the substituted feature had been known as an equivalent before the filing date of the present application in the same technical field.

(Example 1)

In comparison with the prior art, an invention claiming a heat exchanger is characterized by substitution of Sic with Aluminum which has an equivalent property. In this case, the inventive step of the invention is negated if it had been known before the time of filing the application that Sic and Aluminum are equivalents in providing a light and anti-corrodible heat exchanger.

(Example 2)

An invention discloses use of magnesium carbonate to promote crystal formation by quickening a reaction, instead of use of magnesium oxide as in the prior art.

However, if it had been known that magnesium carbonate changes into magnesium oxide when the reaction temperature rises over 1,300°C, the substitution of magnesium carbonate for magnesium oxide is only considered as a replacement with equivalents, thereby nullifying the inventive step of the invention.

(Example 3)

An invention relates to a drill comprising a hydraulic motor, and the prior art relates to a drill comprising an electric motor. At the time of filing the application, the exchangeable use of a hydraulic motor and an electric motor had been very well-known so that a person skilled in the art can hardly expect an unforeseen advantage. In this case, the inventive step of the invention can be negated.

6.2.2 Workshop modification of a design in applying a specific technology

When an invention can be merely arrived at by applying normal design procedures without affecting the technical concept of the prior art, and is neither described as having nor shown to lead to any advantageous effects, the inventive step of the invention cannot be acknowledged.

For example, if the difference between the claimed invention and the cited prior art is only caused by the application of particular parameters such as size, proportion, relative dimensions, and amount from a limited range of possibilities, the inventive step cannot be acknowledged. But if the difference can lead to any particular change in any functions or reactions with an unexpected advantage, the invention can be determined as having an inventive step.

(Example 1)

The claimed invention is provided to prevent the movement of a door of a microwave when a user tries to open or close the door, by installing movement prevention protrusions on the upper and lower sides of the door to engage with insertion grooves, which is different from the prior art in the physical structure, size, numbers, and positions of the protrusions and insertion grooves. The

difference in the configuration can be obtained by a normal design procedure to adopt insertion members and engagement members. Hence, the inventive step can be negated.

(Example 2)

Compared with a microcomputer for a Kimchi refrigerator of the prior art, a control circuit for an electric massager of a claimed invention is only different in the resistance information and operation type of an electric motor which are adjusted for the electric massager. In terms of the design technology of the microcomputer at the filing time of the application, such a difference could be arrived at applying a normal design procedure. Hence, the inventive step is negated.

(Example 3)

The claimed invention is related to a level gage cover for a water tank wrapped by heat insulation. The prior art relates to a door with a sealing material on its inner surface. At the first sight, the present invention is likely to be arrived at by simply replacing the sealing material of the prior art with heat insulation. But adoption of heat insulation for a level gage would prevent the gage cover from freezing and cracking in case of a sudden temperature drop. In this case, the examiner has to carefully consider the effects following the design procedure in assessing the inventive step.

6.2.3 Partial removal of technical features

The claimed invention is not considered as having an inventive step, if, with regard to the state of the art, the omission of some technical features readily mentioned in the prior art causes removal of the related function and effect, and is obvious to a person skilled in the art. But regarding the common knowledge at the time of filing, the inventive step can be acknowledged when the omission of some features does not affect the function of the invention or rather enhances the function.

(Example 1)

The prior art is a toothpaste containing water-soluble silicate, wherein the silicate forms a teeth surface membrane having the effect of protecting sensitive teeth from a stimulus, whereas the claimed invention does not contain said water-soluble silicate to lower the manufacturing costs. At this time, the effect of silicate for coating the surface of teeth and preventing them from stimulus is also removed. For this matter, the present invention is not considered to be inventive.

6.2.4 Simple change and limitation of use of the invention

A claimed invention consisting merely of use of a known invention or in a further simple restriction of such use is not considered inventive. In other words, the claimed invention which is distinguished from the prior art only in a modification of its use or further extension of its use without exhibiting any advantage is not considered inventive.

(Example 1)

A synthetic oil which delays the change of lubricating properties is disclosed in the prior art, whereas the claimed invention discloses reuse of synthetic oil as cutting oil during a cutting process. In this case, if recycling of the synthetic oil as cutting oil is naturally expected from a delayed change of the lubricating properties, the inventive step is negated.

6.2.5 General application of known art

The claimed invention consists merely of use of a known technique in a closely analogous situation in order to solve a problem posed by the prior art with readily anticipated effect, which is not inventive. On the other hand, the claimed invention may be considered inventive when the application of the known technique leads to unexpected beneficial effects in combination with other components in comparison with the prior art.

(Example 1)

The claimed invention is characterized in transforming conventional

Woowhangchungshimwon into liquid type for administration. The inventive step can be negated if such a transformation in the administration type from a solid pill type to a liquid form for oriental medications is within common practice.

(Example 2)

The claimed invention relates to formation of a leak detecting hole in a pipe connecting joint, which is considered not inventive since the technical feature of making a penetrating hole in the outer surface to observe the inner space of an article is commonly practiced without exercising any ingenuity.

(Example 3)

The claimed invention relates to a tray for storing components of a ball grid array integrated circuit, which is not considered inventive if the pin type component for integrated circuit has already begun to be replaced with a ball grid type component at the filing time of the application, since a person skilled in the art would adopt without any special difficulty the ball grid type tray which is prevalent at the filing time of the invention.

6.3 Advantageous effects to be considered

(1) If an effect derived from matters defining a claimed invention is advantageous in comparison with an effect of a cited invention, it is taken into consideration as a fact to affirmatively support its inventive step.

(Reference 1)

Under Article 29, paragraph (2) of the Patent Act, if a claimed invention could have been easily made from the prior art before the filing of a patent application by a person skilled in the art, a patent for such an invention may not be granted. However, when an advantageous effect compared to the cited invention is so remarkable that it could not have been foreseen by a person skilled in the art from the state of the art, there may be cases where the inventive step is not denied (Supreme Court's Decision, 1997. 9. 26. 96Hu825).

(Reference 2)

If a claimed invention is made by collecting and improving commonly known and commonly used arts, the claimed invention is not considered to be inventive except the case where it is difficult to colligate the arts and achieve a new advantageous effect more than that expected from the prior arts, thereby the cited invention could not have been easily made by a person skilled from the prior art, and the case where a new technical method is added to the claimed invention (Supreme Court's Decision 1997. 5. 30. 96Hu221 sentence).

(2) Even if the claimed invention is considered to be easily made by combining cited inventions at the first glance, if the claimed invention has an advantageous effect, qualitatively different or qualitatively the same but quantitatively prominent in comparison with those of the cited inventions, and if the advantageous effect could not have been foreseen by a person skilled in the art from the state of the art, the inventive step can be acknowledged.

Particularly, in the case of an invention in a technical field in which an effect of a product is difficult to predict from its structure like a selection invention and chemical inventions, the advantageous effect compared to the cited invention is an important factor to positively infer the inventive step.

(Reference)

The inventive step of a composition of dyes comprising more than two chemical compounds mixed in a certain ratio should be determined over the case when each of the chemical compounds solves the problem. Even though individual elements of the chemical compounds belong to different categories, the claimed invention has an inventive step if mixing the chemical compounds in a certain ratio leads to unexpected results (Supreme Court's Decision 1994. 4. 15. 90Hu1567).

(3) Though the advantageous effect of the claimed invention is superior to that of the prior art and is not explicitly disclosed in the detailed description, the examiner can assess the inventive step from the inventor's assertion and evidence like experiment results, if the effect is easily recognized by a person skilled in the art from the detailed description and the structure of the invention shown in

the drawings. But if such an assertion is not supported by the detailed description and is not inferred from the description or drawings, the effect from the assertion should not be considered in assessing the inventive step.

(Example 1)

The claimed invention relates to a blood-cupping device characterized by a half-open barrel extendably installed on the lower part of an operation stick, which is aimed at easily checking the movable rubber plate inserted in the barrel and facilitating the airflow through the half-open barrel while removing of the main cup body. In this case, if the effects of the half-open barrel are remarkably ensured from the description and common knowledge by a person skilled in the art, the claimed invention can be considered inventive.

6.4 Determining the Inventive Step According to the Invention Type

6.4.1 Determining the Inventive Step of a Selection Invention

A selection invention is an invention with a more specific concept selected from a generic concept disclosed in a cited invention, which includes a composition which is not directly disclosed in the cited invention as an indispensable part.

If in a claimed invention, an optimized concept is experimentally selected from publicly known technology, the inventive step of the claimed invention cannot be acknowledged because selecting the best or suitable concept from publicly known technology comes within the scope of an exercise of ordinary creativity of a person skilled in the art. However, if a selection invention generates advantageous effect in comparison with a cited invention, the inventive step of the selection invention can be acknowledged. In this case, all specific concepts included in the selection invention should have advantageous effects qualitatively different or qualitatively the same but quantitatively prominent. The detail description of the selection invention should precisely explain that the invention generates an advantageous effect in comparison with the cited invention, and needs not provide experimental materials to confirm the prominence of the effect. If the invention is rejected because of the effect is doubted, the applicant can

assert the effect concretely by submitting materials relating to experimental comparisons.

(Example 1)

Both a claimed invention and a cited invention relate to a chemical compound for protecting a nerve, which is used for curing a regressive disease of the central nervous system. If a chemical compound is selected in the claimed invention selects with a more specific concept which is not directly disclosed in the cited invention, and the oral activity of the claimed invention generates ten times more advantageous effects than the cited invention, the inventive step of the claimed invention can be acknowledged.

6.4.2 Determining the Inventive Step of an Invention including Numerical Limitations

An invention with numerical limitation defines that some parts of indispensable elements of the invention described in the claims are expressed by specific numerical values.

Experimentally selecting an optimal numerical range from the publicly known art is normally considered as an exercise of ordinary creativity of a person skilled in the art, and hence the inventive step is generally denied. However, a claimed invention has an inventive step if within a limited numerical range has more advantageous effect than the effect of the cited invention. This advantageous effect should be a remarkably improved effect regarding the overall scope of the numerical limitation. Also, determining the significance of the critical range of the numerical limitation comes under the following criteria.

- (1) The significance of the critical range of a numerical limitation is required with regard to any part of the numerical range if a claimed invention is a continuation of the cited invention.
- (2) If the two inventions each have different problems to be solved and qualitatively different effects, the significance of the critical range of the numerical limitation is not required even though the two inventions have the

same matters defining the inventions except for the numerical limitation.

For the significance of the critical range of the numerical limitation to be acknowledged, a remarkable change in an invention effect is required across the boundary of the numerical limitation and also the following conditions should be satisfied: 1) the technical meaning of the numerical limitation should be described in detail, and 2) embodiments in the detail description or supplemental materials should prove that the range of the numerical limitation is critical. Generally, it should be objectively confirmed that the range is critical with experimental results which cover all range of the numerical limitation.

(Example 1)

The claimed invention discloses a numerical limitation in that the spiral, comprising that one rotation of a screw limits to the tube length which is 12 times longer than the inner diameter. However, since there is no technical explanation of limiting 12 times longer than the inner diameter in the description, it only means that the spiral of a screw is merely not too much gentle and there is no special effects. Therefore, the numerical limitation of the claimed invention is considered to be technically meaningless.

(Example 2)

The claimed invention relates to a ceramic backside material for arc welding to make back bead shape better. The technical difference of the claimed invention is a backside material comprising 0.01 - 0.7% iron-oxide. If this numerical value is merely a numerical limitation that a person skilled in the art can reach by an exercise of ordinary creativity and there is no heterogeneous remarkable effect within the range of this numerical limitation, the claimed invention is not considered to have an inventive step.

(Example 3)

Even though the manufacturing ingredients or process of the claimed invention is similar in some part or identical to those of the cited inventions, if the claimed invention is different from the cited inventions in view of its characteristics such as additives in processing or a ratio of manufacturing ingredients and thus the

quality and economical efficiency of the complete goods are greatly improved, the claimed invention is considered to have an inventive step(Supreme Court's Decision, 1992. 5. 12. 91Hu1298).

(Reference)

If a claimed invention defines the range of technical elements of a known cited invention with numerical values, no other technical elements to prove an inventive step are added, and the numerical limitation is merely a supplemental material, and if there is no remarkable effect within the range of the numerical limitation, the claimed invention is merely a numerical limitation within the scope of the common practice of a person skilled in the art. In other words, if the claimed invention and the cited invention have the same problems to be solved and are different only in the limited numerical values, and if there is no mention in the description about remarkable effects in employing the limited numerical values, it is difficult to admit there is a remarkable effect within the range of the limited numerical values.

6.4.3 Determining the Inventive Step of a Parameter Invention

(1) A parameter invention is an invention in which an applicant creates a certain parameter which is not standard nor commonly used for a characteristic value in physics or chemistry, parameterizes it mathematically by using a correlation between a plurality of parameters, and employs it as a part of essential elements of the invention. Since the parameter invention may not precisely define the subject matter via the claim description, determining the inventive step of the parameter invention should be performed only after figuring out the subject matter based on the detailed description, drawings, and common knowledge.

(2) As the functions and characteristics described in the claim define the subject matter of an invention, the examiner should not compare the claimed invention with the cited invention without considering the functions and characteristics. In case of a parameter invention, the inventive step should be determined by taking into account the functions or characteristics caused by a parameter. For determining the inventive step of a parameter invention, it should be firstly

considered whether a technical meaning exists in introducing a parameter. If the parameter described in claims is merely a matter of expression form different from a publicly known invention or a matter of confirming the intrinsic features of a publicly known invention, and if the cause and effect relationship between the parameter and the advantageous effect is weak, the inventive step of the parameter invention is denied. However, if the parameter invention is a type of an invention with a numerical limitation, the determination criteria of the inventive step of a numerical limitation invention can be applied. In this case, even without the technical meaning of the parameter, as long as an effect of the claimed invention caused by the numerical limitation is considered remarkable, the inventive step of the parameter can be acknowledged.

(3) Although it is difficult to figure out or convert a certain parameter in a claim, and therefore, it is hard to compare the claimed invention with the cited invention, if there is a reasonable doubt that the parameter invention can be easily derived from the cited invention, the examiner can notify the applicant of the reasons for refusal due to the inventive step without having to strictly compare the claimed invention with the cited invention and wait for the applicant's proof statement. If the examiner has difficulty in maintaining the reasons for refusal due to the applicant's refutation, the reasons for refusal are cancelled. If the reasons for refusal are not overcome by the applicant's argument, the examiner may make a decision of refusal under Article 29, paragraph (2).

(4) The examiner might have reasonable doubt in the following cases: ① the parameter described in the claims is obtained from a different definition and measurement method, and then the claimed invention is found to be easily derived from the cited invention. ② the examiner evaluates the parameter of the cited invention according to the measurement method in the description, and then the claimed invention is proved to be similar to the cited invention. ③ an embodiment in the description of the claimed invention is similar or identical to that of the cited invention.

(5) If the examiner notifies the applicant of the reasons for refusal with regard

to a parameter invention, the examiner has to concretely describe the grounds of reasonable doubt, and if necessary, the examiner can propose a solution to overcome the refusal grounds.

(6) If the parameter of the claimed invention is standard, commonly used or proved to be easily understandable by a person skilled in the art, the examination criteria described in (1)-(5) are not applied.

(Reference)

In comparing the claimed invention having certain properties or characteristics with the cited invention having different properties or characteristics, if the claimed invention becomes similar or identical to the cited invention as a result of converting the properties or characteristics of the claimed invention with different definitions and measurement methods, or if an embodiment of the claimed invention in the description is similar or identical to one of the cited invention, the claimed invention is not considered to be novel and have an inventive step because the two inventions should be considered to be similar or identical to each other (Supreme Court's Decision 2002. 6.28 2001Hu2658)

6.4.4 Determining the Inventive Step of a Product Invention described by its manufacturing process

As an applicant should specifically describe the article composition of the invention when defining a product invention, even though a manufacturing method is described in the claims of the product invention, without any exception, the examiner can determine the inventive step of the product invention by comparing the product itself defined by the description with a publicly known invention without considering the manufacturing method.

In determining the novelty and inventive step, as the object to be protected, is not the manufacturing method but the product itself in the product invention described by its manufacturing method, the examiner shall compare the composition of the product itself with the composition of the publicly known product. The examiner does not have to take into account the manufacturing

method or manufacturing apparatus for the product. In this case, the object of determination is a product characterized by certain properties, features, and composition based on the method description.

(Example 1)

The claimed invention is about a belt coupling apparatus for a seat belt and is described in the claims with a manufacturing method, comprising the steps of: bending a part of a plate from one lateral side to other lateral side and concurrently putting the bent part back to the first lateral side. In this case, as there is no difficulty in characterizing the seat belt via its composition, the examiner can determine the inventive step by comparing the seat belt defined by the manufacturing method with the cited invention, without considering the manufacturing method.

(Example 2)

The claimed invention relates to a method of producing cannaf tea, and it is described in the claims that the available components are the cannaf leaves whose inorganic content is increased by heating (60 C, 45min.) and infrared radiation (1.6 kw, 30-45 min.). In this case, if it is proved, based on the description of the specification, that the inorganic content of the cannaf leaves produced by the described method is remarkably increased, the inventive step of the cannaf tea produced by the described method is acknowledged because of the change of the tea properties.

7. Determination of the inventive step of a combination invention

(1) A combination invention is an invention comprising novel solutions by gathering technical features disclosed in the prior art as a whole in order to solve a technical problem.

The invention described in a claim is to be considered as a whole. Accordingly, the inventive step of the combination invention shall not be negated merely because each element described in a claim is deemed to be known from or obvious over the cited inventions.

That is, in the case of a claim disclosing a plurality of elements, determining the inventive step relies not upon each independent element, but upon the technical idea of the claimed invention, the respective elements of which are structurally combined as a whole. Therefore, when determining the inventive step is, the examiner shall consider the difficulty in forming structurally combined elements as a whole based on the principle of a problem solution, rather than consider whether individually dissected elements in the claim are publicly known. In addition, the examiner shall consider the unique effect that the invention has as a whole.

(2) Determining the inventive step of the combination invention can be made by combining more than two disclosures (well-known or commonly used art) but the combination of the disclosures is limited to the condition where a person skilled in the art can easily combine the disclosures at the time of filing. In this case, there is no special limit on the number of prior art disclosures to be combined. When the examiner determines the inventive step by combining various prior arts, the examiner mainly considers whether the cited inventions contain a motivation or hint leading to the claimed invention by combining or assembling the prior art disclosures. Nevertheless, taken into account the state of the art, the common general knowledge at the time of filing, the general technical problems of the technical field, the technical trend and demands in the industry, if the combination of prior art disclosure is deemed to be easily made by a person skilled in the art, the examiner can deny the inventive step of the claimed invention (Supreme Court's Decision 2007. 9. 6. 2005Hu3284).

(Reference)

Well-known art means technologies generally known in the relevant technical field like technologies widely known throughout the industry, technologies that appeared in many prior art disclosures, or technologies well known to the extent to present examples. Commonly-used art means well-known art which is used widely.

(Example 1)

The claimed invention relates to a web game server enabling users to download a game via the Web. The technical feature of the claimed invention differs from that of the cited invention only in that a game program and game data are separately downloaded in the claimed invention. In this case, if the technical difference in the game program and game data separately downloaded is deemed to have been merely a well-known art in view of the state of the art at the time of filing, the inventive step of the claimed invention is not acknowledged, as a person skilled in the art could have simply combined it with the cited invention without any difficulty.

(Example 2)

The claimed invention relates to a method of counting securities by extracting serial numbers via an image sensor. Compared to the cited inventions, the claimed invention differs from cited invention 1 only in that the cited invention 1 recognizes security denominations via an optical sensor, and cited invention 2 comprises the step of sorting currency notes via an image sensor. Considering the state of the art at the time of filing and the fact that the prior art disclosures fall under the same technical field, the difference between the invention sought to be patented and the prior art would have been obtained by substituting the image sensor of the cited invention 2 for the optical sensor of the cited invention 1 without difficulty. Therefore, it would have been obvious to a person skilled in the art to combine the cited inventions 1 and 2, thereby arriving at the claimed invention.

(3) The determination whether a prior art disclosure contains a motivation, hint, or the like for a combination shall be made by synthetically assessing the following: whether the motivation, hint, or the like is explicitly taught in the prior art; whether the motivation, hint, or the like is inherent from the technical problem to be solved by the invention; or whether the motivation, hint, or the like is part of the common general knowledge or empirical rules of a person skilled in the art.

(Example 1)

Cited invention 1 discloses a protective cover of a baby carriage comprising a

transparent window made of a flexible plastic material, while the claimed invention describes a protective cover with a transparent window whose material is changed to a rigid plastic material disclosed in cited invention 2 in order to protect the eyesight of an infant. In this case, if a public TV program reported that a flexible plastic material used for the transparent window of a baby carriage damaged the eyesight of an infant prior to the priority date and if the fact that a rigid plastic material did not result in such a problem fell under the common general knowledge of the art to which the invention pertains, a person skilled in the art could have changed the material of the transparent window disclosed in the cited invention 1 to the rigid plastic material of the cited invention 2 without any difficulty. Therefore, the claimed invention would have been obvious to a person skilled in the art.

(4) In general, as a prior art disclosure referring to another disclosure can be considered to explicitly suggest a hint or motivation of a combination in the prior art disclosure, it is regarded as obvious to combine the two disclosures and the inventive step is therefore negated. Also, combining a plurality of technical features in the same disclosure is considered obvious, for a person skilled in the art would have combined the technical features without difficulty.

It is normally considered to be obvious to combine a well-known technology with another prior art disclosure. However, if a technical feature to be combined is a well-known technology in the art, but a combination with another technical feature results in an advantageous effect, the combination is not regarded as obvious.

(Example 1)

If cited invention 1 discloses all elements except for a leading portion of the claimed invention, and the leading portion of the claimed invention is substantially the same as the guide member of cited invention 2 referred to in cited invention 1, it would have been obvious to combine the cited invention 2 with the cited invention 1, thereby arriving at the claimed invention because the combination of the cited inventions 1 and 2 can be considered as being already implied.

(5) In general, if a combination invention achieves an effect by functional interaction between technical features, which is different from or greater than the sum of the effects of the individual technical features, e.g., a combined synergistic effect, the inventive step may be acknowledged since a set of technical features is considered to be a technically meaningful combination. If a combination invention described in a claim is regarded not as a meaningful combination, but merely as a juxtaposition (array) or aggregation (simple collection) of features, the inventive step of the combination invention may be denied by proving that the individual features are obvious insofar as there are no other grounds supporting the inventive step.

(Example 1)

The claimed invention is similar to the cited invention 1 except for a servo motor modified from a hydraulic actuator of cited invention 1 and a bending means described in cited invention 2 and substituted for the spindle of cited invention 1. In this case, if the modification or substitution of the elements does not lead to structural difficulties and the functional effect of new elements is not regarded as greater than the summed effects of the cited invention 1 and the cited invention 2, the claimed invention falls within an aggregation, and is therefore denied inventive step.

(Example 2)

The claimed invention corresponds to an aggregation of an ordinary injection molding machine disclosed in cited invention 1, a vacuum chamber disclosed in cited invention 2, and a mold fastening system disclosed in cited invention 3, wherein the vacuum chamber enables injection molding to be performed in vacuum and the mold fastening system facilitates work convenience. In this case, if the combination of the elements does not lead to particular difficulties, nor does the functional effect result in any remarkable difference, the aggregation is considered to be obvious to a person skilled in the art, thereby arriving at the claimed invention.

(6) In determining the inventive step of a combination invention, care must be

taken as the fact that one or more cited inventions must be combined with the closest cited invention in order to arrive at the claimed invention may indicate the presence of an inventive step. Also, it should be noted that the higher number of combined cited inventions, the more likely is that the claimed invention results from an ex post facto view or lacks a valid reason for rejection. When determining whether it would have been obvious to combine two or more other prior arts, the examiner should take into consideration of the followings: ① whether there is good possibility to combine them, ② whether the prior arts come from similar or neighboring technical fields, and ③ whether there is a reasonable basis to associate each other for the combination.

8. Other factors to be taken into account in determining obviousness

In principle, the determination of the inventive step is to consider synthetically the objective, technical constitution, and functional effect of an invention described in a claim, i.e., to determine the uniqueness of the objective and the remarkableness of the effect as a whole, mainly based on the difficulty of technical structure. However, there might be other factors in determining the inventive step. Thus, the examiner should not readily reach the conclusion that the claimed invention lacks an inventive step if a written opinion submitted by an applicant claims that the claimed invention is not obvious for the following reasons:

(1) If a prior art document teaches not referring to the prior art thereof, i.e., if there is a description in the prior art document that precludes the reasoning that a person skilled in the art would easily arrive at the claimed invention, the inventive step is not denied by the prior art despite the similarity between the prior art and the claimed invention. In addition, the fact that the prior art in a prior art document is described as inferior cannot be necessarily considered as a factor that precludes the inventive step.

(2) Commercial success or favorable comments from the industry or the fact that the claimed invention had not been implemented by anybody for a long time before the claimed invention was filed may be regarded as indicative of the

inventive step as secondary evidence. However, those facts alone are not to be regarded as indicative of the inventive step. First of all, as the inventive step should be determined based on the contents disclosed in the specification (i.e., the objective, structure, and effect of the invention), commercial success is not to be regarded as a reference for the determination of the inventive step, provided that such success does not derive from the technical features of the invention but from other factors (e.g., improvement in sales techniques or advertising).

(Example 1)

Although a mobile video pop song accompaniment of the claimed invention made a hit in Japan with a signed two-year export contract worth \$84,000,000, this cannot prove that the success is based only on the superiority of a technical structure of the claimed invention. In addition, if the success is determined as deriving from the sales techniques of a salesperson, evidence of the commercial success alone is not to be regarded as a factor in guaranteeing the inventive step.

(Example 2)

The claimed invention is related to a method of fixing metal accessories on a handrail, wherein a welding hole and a curved surface each have a size appropriate for welding so that internal welding can be performed. If the claimed invention had a better functional effect than a connection apparatus of a handrail pillar of prior external welding but the claimed invention had not been implemented before filing, then it would be regarded as non-obvious for a person skilled in the art to arrive at the claimed invention.

(Reference)

Given that the claimed device has a distinguished functional effect but has not been implemented before filing, it is deemed to be highly non-obvious to a person skilled in the art to devise the claimed device of this case (Precedent Case 99Hu1140).

(3) The fact that a claimed invention solves a technical problem that a person

skilled in the art have attempted to solve for a long time or fulfills a long-felt need may be regarded as an indication of the inventive step. In addition, such a solution of a technical problem or a need should be fulfilled by the claimed invention for the first time as a matter that has been recognized by a person skilled in the art for a long time. To accept this as an indication of inventive step, objective evidence is required.

(4) If an invention is made by employing technical means which a person skilled in the art has abandoned due to technical prejudice interfering with the research and development of a technical problem in the relevant field of the art, thereby solving the technical problem, this is regarded as an indicator of the inventive step.

(5) If a claimed invention proposes means for overcoming technical difficulties not resolvable by other means or for solving a technical problem, this is regarded as advantageous evidence for an inventive step.

(6) If a claimed invention falls within the area of a brand-new technology and has no prior art relevant to the invention, or if the closest prior art to the invention is far away from the invention, the inventive step is likely to be acknowledged.

9. Notes on determination as to whether the claimed invention has an inventive step

(1) When determining the inventive step in the light of knowledge obtained from the matters disclosed in the specification of a patent application, which are the subject of examination, it should be noted that the examiner can rapidly conclude that it would have been obvious for a person skilled in the art to arrive at the invention described in the claims without difficulty.

(Example 1)

If the claimed invention is related to a terminal for use in an emergency situation, comprising a tapping mode blocking a voice signal reception of a

receiving unit and only allowing a voice emission of a transmitting unit and if the cited invention merely discloses 'preventing others from noticing a voice reception from a terminal', the specific technical structure of the claimed invention is not obvious to a person skilled in the art. Therefore, the assessment that the claimed invention would have been obviously derivable from the description of the cited invention above results from an ex post facto analysis on the basis that the examiner knows the contents disclosed in the specification.

(2) If an independent claim has an inventive step, a claim dependant on the independent claim is deemed to have an inventive step too. However, if an independent claim does not have an inventive step, the determination should be made for each claim dependant on the independent claim.

(3) If a product invention explicitly has an inventive step, a process invention for making the product or a use invention for using the product has also an inventive step in principle.

(4) For a claim described in a Markush form or comprising multiple selective technical features, if at least one invention of the Markush alternatives is proved not to have an inventive step based on the prior art, the applicant can be notified of the reason for refusal for the claim. In this case, the applicant can overcome the reasons for refusal by deleting the Markush alternatives which do not have an inventive step. On the other hand, in determining the inventive step of a claim described in a Markush form or comprising multiple selective technical features, care must be taken not to expand the effect of one of the selective elements to the whole effect of the claimed invention.

(Example 1)

If the claimed invention relates to neuroprotective chromanol compounds including various chemical compounds as selective elements, all of the chemical compounds must have a remarkable effect over one or more cited inventions in order for the claimed invention to be granted. Thus, it is not correct for an examiner to grant a patent based only on comparison test data concerning an alternative (formula (III) compound) described in the specification as being

significantly effective.

(5) A degenerate invention does not involve an inventive step. Granting a patent to a degenerate invention runs against the purpose of the Patent Act. Moreover, a degenerate invention is barely conducted and it would rather have a negative effect on those who use it, even if the degenerate invention is given exclusive rights by granting a patent.

(6) If the cited invention is regarded as a well-known technology, then the examiner may notify the applicant of the reasons for refusal without any evidential material attached. However, it is not appropriate to regard a well-known technology as being the closest cited invention without any evidential material.

In response to the reasons for refusal on the basis of a well-known technology without any evidential material attached, if an applicant claims that the invention is not a well-known technology in a written opinion, the examiner should in principle provide an evidential material with regard to the reasons for refusal. If the examiner has difficulties in providing an evidential material, the examiner may deny the inventive step by fully explaining why the invention falls under a well-known technology or pointing out why the applicant's argument is not proper (the evidential materials with regard to a well-known technology may be widely-used textbooks, introductory books, technical standards dictionaries, national standards (KS) in the field of the art to which the matter pertains, and so forth).

(7) As the inventive step of an invention is individually determined, not based on examination of precedents of other inventions, in the case of a patent application for a specific invention, examination precedents in foreign countries that differ in terms of the legal systems and customs are not taken into consideration.

(8) Although the implementation of a technical content of the claimed invention is prohibited due to the law restriction in Korea and abroad, such a restriction is

not taken into consideration in determining the inventive step.

(Example 1)

If the claimed invention and the cited invention differ only in a method of lottery drawing and the method is strictly prohibited by law and cannot be readily modified by the lottery's designer, the claimed invention is obvious to a person skilled in view of its technical difficulty only and is considered to have no inventive step, for the law restriction is not taken into consideration.

Chapter 4. Enlarged Concept of Novelty

1. Patent Act 29 paragraph (3)

Notwithstanding Patent Act 29 paragraph (1), where an invention claimed in a patent application is identical with an invention or device described in the specification or drawing(s) originally attached to another application for a patent or for a registration of a utility model, which was filed prior to the date of filing of said patent application and has already been laid open or published after the filing of said patent application, the patent may not be granted for such an invention. However, this may not apply, where the inventor of the concerned patent application and the inventor of the another application for a patent or for a registration of a utility model are the same person, or where the applicant of the concerned patent application and the applicant of the another application for a patent or for a registration of a utility model are the same person at the time of the filing of the concerned patent application.

2. Purport

For an applicant an invention described in a specification or drawings, if not in claims, which is usually opened to the public by a laying-open of application or publication of registration, has been contributed to the society without reward. Giving exclusive rights to another applicant filing a subsequent application which claim the invention which belong to the public is inconsistent with the purpose of the Patent Act, which gives exclusive rights for designated periods to new inventions as a reward for the disclosure. Therefore, granting a patent to such subsequent invention is not allowed under the Article 29 paragraph (3), (4).

Moreover, the Article 29 paragraph (3) and (4) aim to prevent an unexpected problem below, where an applicant discloses the invention, which is only described in specifications and drawings, in claims by the amendment, the application can possibly be a prior application under the Article 36 of the Patent Act, in turn conducting a subsequent application would be delayed by the end of the examination of the prior application.

3. Conditions to meet the requirement of the Article 29 paragraph (3) and (4)

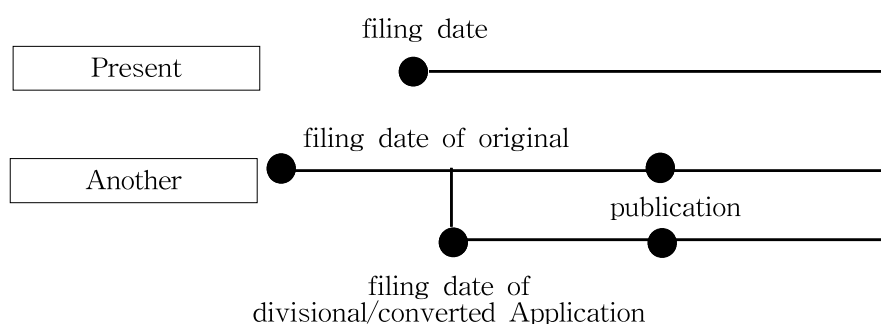
In order to satisfy the requirements of Article 29 paragraph (3) and (4) of the Patent Act, the following conditions must be met.

(1) Another application for a patent or for a registration of a utility model cited as a reference under the Article 29 paragraph (3) and (4) (referred to as "another application" hereinafter) is required to have a filing date prior to the filing date (the filing date of the first filing country in case of the application with priority claim under the Paris Convention or the filing date of prior application in case of the application with domestic priority claim) of the concerned patent application (referred to as "present application" hereinafter).

① In the case where another application is a divisional application or a converted application (double application in case of the application filed prior to 1 October 2006.) under the Article 29 paragraph (3) and (4), the critical date of filing as a reference is the actual filing date of a divisional application or a converted application, not the date of filing of the original application.

(Example 1)

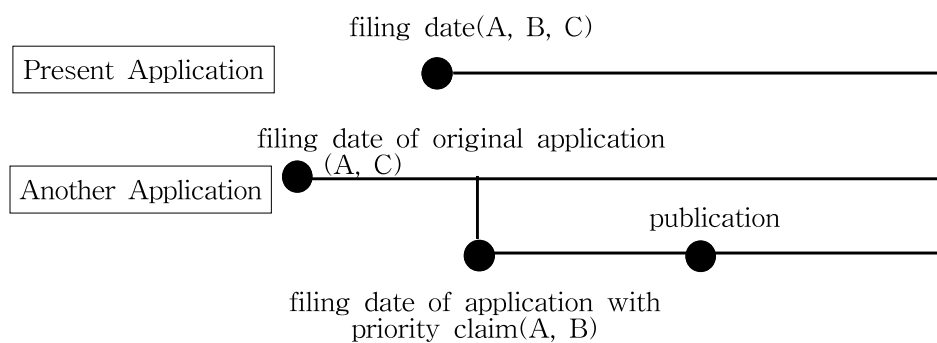
When applying Article 29 paragraph (3), if the filing date of such application is after the filing date of the present patent application, the divisional application or a converted application cannot be cited as a prior art. However, the original application of such application can be another application and be cited as prior art if the filing date of the original application is prior to that of the present patent application (see figure below).



② In the case where another application is one with a priority claim under the Paris Convention, the filing date in the country of origin is deemed as the filing date of another application, for an invention commonly disclosed in the specification or drawings(referred to as an "specification, etc." hereinafter) attached to the application in the country of origin and in the specification or drawings originally attached to the application with the priority claim.

(Example 1)

The invention A of another application can be cited as a prior art of the present patent application, if the filing date of invention A disclosed in the application filed in the country of origin as a reference is deemed to the date of filing in the country of origin when applying Article 29 paragraph (3). The invention B cannot be cited as a prior art if the invention B was not disclosed in the application filed in the country of origin. Meanwhile, the invention C, which is described in the application filed in the country of origin but is not included in the present application with priority claim, was not disclosed in the application filed in Republic of Korea cannot be cited as a prior art of the present patent application.



③ In the case where an examiner cites an invention initially disclosed in a specification of a prior application which was a basis for a domestic priority claim or an application with a domestic priority claim thereof (referred to as a "subsequent application" hereinafter) as an invention of another application under the Article 29 paragraph (3) and (4), it is treated as follows:

(a) The invention commonly disclosed in the initial specification of both prior and subsequent applications, is deemed as another application filed on the filing date of prior application and should be applied in the provision of Patent Act Article 29 paragraph (3) and (4) (Patent Act Article 55 paragraph (3) and (4)). The invention solely disclosed in the initial specification, etc. of a subsequent application but not in that of prior application, is deemed as another application filed on the filing date of subsequent application and should be applied in the provision of Patent Act Article 29 paragraph (3) and (4) (Patent Act Article 55 paragraph (4)). The invention solely disclosed in an initial specification, etc. of prior application but not in that of subsequent application should not be applied in the provision of Patent Act Article 29 paragraph (3) and (4) (Patent Act Article 55 paragraph (4)).

An prior application is deemed to have been withdrawn when more than one year and three months (in the case of the application for a registration of a utility model filed after 2001. 7. 1, immediately applied) has elapsed after the filing date of the prior application (Patent Act Article 56 paragraph (1)) and is not laid open. Therefore, where a subsequent application is laid open or published, the invention commonly disclosed in the initial specification of both prior and subsequent applications is deemed as laying-open at the time for lay-open or publication of subsequent application. In addition, where an invention was not disclosed in the initial specification of the prior or subsequent applications but newly described through the amendment, the treatment above does not apply. Where an invention was not described in the initial specification of subsequent application but described in the initial specification of prior application, the invention is not deemed to be laid open to the public. Therefore, the Article 29 paragraph (3) and (4) are not applied to such an invention.

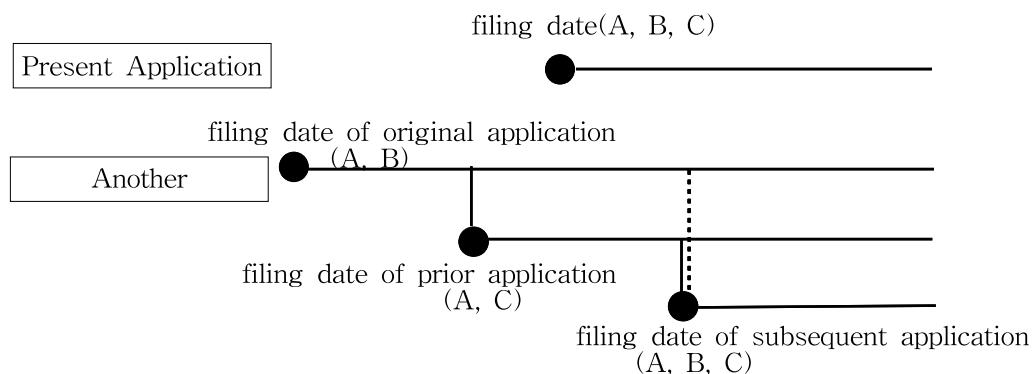
(b) In the case of (a), where a prior application is an application with a domestic priority claim (including a priority claim under the Paris Convention), the invention commonly disclosed in the specification of both prior and subsequent applications is deemed as another application filed on the filing date of subsequent application and should be applied in the provision of Patent Act Article 29 paragraph (3) and (4) (Patent Act Article 55 paragraph (5)).

(Example 1)

As example ① shows below, where a subsequent application is filed with a priority claim based only on a prior application, among the inventions A and C described in the prior application, the invention A disclosed in the original application filed at the country of origin, which is a basis of the priority claim of the prior application under the Paris Convention, is considered to be filed at the filing date of subsequent application under Article 29 paragraph (3). Therefore, invention A of the subsequent application cannot be cited as a reference under Article 29 paragraph (3) even though the invention A is described in the prior application. The invention C of the prior application can only be cited as a reference.

(Example 2)

Meanwhile, as example ② shows below, where a subsequent application was filed with priority claims based on the application filed at the country of origin as well as a prior application, the invention A of the prior application can be cited as a reference under Article 29 paragraph (3).



(Footnote)

Case ① : In the case where a subsequent application is filed with a priority claim based only on a prior application

Case ② : In the case where a subsequent application is filed with priority claims based on the application filed at the country of origin as well as a prior application

(2) Another application should be laid open or published for registration after the present application was filed.

Once an application is laid open or published for registration, the status of the application as another application still remains effective despite refusal or grant of a patent, invalidation, withdrawal or abandonment of the application. However, in the case where an application is laid open to the public, after the decision of refusal, invalidation, withdrawal and abandonment of the application, the application cannot be cited as a reference under Article 29 paragraph (3).

(3) An invention described in the claims of the present application should be identical to an invention or device described in the initial specification of another application

The invention described in the claims of the present application should be completely or substantially identical to an invention or device described in the initial specification of another application. In addition, even if the matters which have been described in the initial specification of another application are omitted by the amendment after the filing, Article 29 paragraph (3) is implemented.

4. Guidelines applying the proviso of the Article 29 paragraph (3)

An application falling within the following conditions are not considered as another application under Article 29 paragraph (3) and (4).

(1) In a case where an inventor of the present application is the same as the inventor of another application

「The inventor」 of the present application and another application means the inventor described in the application document. In case of joint inventions, all inventors of present application must be completely identical with those of another application. However, even if all inventors do not completely correspond, if the applicant proves the fact that all inventors are practically the same,

examiner can admit the applicant's argument. In a case where an applicant adds or amends an inventor of an application after an examiner notifies the applicant of the reasons for refusal, in which another application is cited as a reference due to the difference of the inventor, he or she has a right to request documents to testify that the inventors added or amended are true inventors.

(2) In the case where the applicant of the present application at the time of filing is the same as the applicant of another application

The applicants of the present and another applications should be identified and compared at the time of filing the present application to determine whether the applicants are the same. In the case of plural applicants, all of the applicants indicated in the two applications must be the same. Even in the case of the subsequent discrepancy of applicants caused by the change of name, inheritance or a merger of applicants of the present and another applications between the filing date of the present application and that of another application, the sameness of applicants remains effective only if the applicants of the present and another applications are substantially same.

5. In the case of international application under Article 29 paragraph (4)

Where another application is an international application or international application considered to be patent application by decision, the following points are different when applying Article 29 paragraph (3), compared to another application which is not an international application.

In a case where another applicant is an international application, in Article 29 paragraph (3), "laid open" refers to "laid open or international publication under PCT Article 21" and "an invention or device described in specification or drawings initially attached to the application on the filing date" means "an invention or device commonly described in description, claims and drawings submitted at the date of filing of the international application and its translation". Under Article 29 paragraph (3) and (4) in the international application with the domestic priority claim, it is considered that the invention described in the specification or drawings of prior application being the basis of the domestic

priority claim among inventions commonly described in description, claims and drawings submitted at the date of filing of the international application and its translation, is considered to be laid open to the public at the time of international publication under PCT Article 21 or publication of registration(Article 202 paragraph (2)).

Under Article 29 paragraph (4), where another application is an international application, the scope of applying enlarged concept of novelty of the international application includes the invention commonly described in description, claims and drawings submitted at the date of filing of the international application and its translation. However, the present application shall be refused by an invention described in the international application (Article 55 paragraph (4)) in the case where the international patent application is a prior application and is the basis of domestic priority claim for another application, if the subject matter of the present application is the same as the invention described in the description, claims and drawings submitted at the date of filing of the international application,.

6. Method of determining the identity

Determining of applying Article 29 paragraph (3) and (4) refers to whether an invention described in the claims of the present invention is identical to the invention or device specified in the description or drawings in another application at its filing (hereinafter referred to as "cited invention").

6.1 Procedure of determining the identity

(1) Find an invention of the present application described in the claims . The method of finding invention is identical to that of determining novelty in chapter 2.

(2) The subject matters described in description of another application should be reviewed. The subject matters which can be obtained by the common general knowledge might be the basis of a cited invention under Article 29 paragraph (3) and (4).

(3) Find identity and difference by comparing the claimed invention of the present invention with cited invention. In this case, the claimed invention should not be compared with the invention by combining more than two cited inventions.

(4) Where there is found no difference between the subject matters defining the claimed invention and the subject matters defining the cited inventions, the claimed and cited inventions are identical. In this case, the identity of the inventions includes the substantial identity.

6.2 Substantial method of determining the identity

Novelty is determined by comparing whether a present invention and a cited invention are identical. The identity of invention is related with determining novelty as well as inventive step (Article 29 paragraph (2)), invention not considered to be publically known (Article 30), enlarged concept of novelty (Article 29 paragraph (3), (4)), protection of lawful holder of a right (Article 33, 34), prior application (Article 36), succession to the right to obtain a patent (Article 38 paragraph (2), (3), (4)), divisional application (Article 52), converted application (Article 53) and application with the priority claim (Article 54, 55). Therefore, the criteria of determining the identity of invention shall apply in the cases mentioned above.

(1) Determining identity of the inventions relies on identity and difference with comparison between the subject matters defining the claimed invention and the subject matters defining the cited inventions.

(2) In a case where there is a difference between the subject matters of the claimed invention and the subject matters of the cited inventions, two inventions are not identical. Meanwhile, if there is no difference between them, the invention in the claims is identical to a cited invention.

(3) In the case where the claimed invention is completely or substantially

identical to a cited invention, the claimed invention is identical to a cited invention.

Chapter 6 Unpatentable Invention

1. Patent Act Article 32

Notwithstanding Article 29 paragraph (1) to (2), an invention likely to contravene public order or morality or to injure public health may not be patented.

2. Purport

For reasons of public interest, Article 32 stipulates that a patent would not be granted for 「inventions likely to contravene public order or morality or to injure public health」 even if the invention falls within patentable subject matter of Article 29 paragraph (1) to (2). As a result, a patent cannot be granted for an invention which falls under the Article 32 without having to consider patentability requirements under Article 29.

3. Unpatentable invention

3.1 Invention likely to contravene public order or morality

An invention likely to contravene public order or morality is considered as unpatentable. In general, the two words are not separately used but more specifically, public order refers to the general interest of society or country and morality means moral sense generally accepted by a society or by a particular group of people. Therefore, it is no wonder that a patent cannot be granted for an invention likely to contravene public order or morality without having to consider the objective of the Patent Act.

An invention which falls within subject matters under Article 32 is an invention with the aim to contravene public order or morality as well as the disclosure or its use is against public order or morality. But it does not extend the case where an invention is likely to go against public order or morality as a result of improper use of the invention against its original purpose.

For example, in a case where an apparatus (Bingo) of a present invention is aimed at entertaining not gambling or other gambling behaviour clearly disclosed in the description and furthermore it is considered that the apparatus is likely to be devised for entertainment use not wrongdoing, the apparatus would not oppose public order or morality just because of the reason that an apparatus could be improperly used in wrongdoing.

3.2 Invention likely to injure public health

An invention likely to injure public health is equally treated as an invention to contravene public order or morality aforementioned. The determination of whether an invention harms the public health should be made in the same way that whether the invention contravenes public order and morality is determined.

In a case where a present invention is manufacturing process invention, an examiner should consider whether the manufacturing process itself and the product made by the manufacturing process would injure public health.

Even in the case where researches of academic journals show that a product made by a manufacturing process of present invention would injure public health, if the Ministry of Health and Welfare approves the manufacture of medicines under Pharmaceutical Affairs Act, it does not mean that the invention can injure public health due to the research results of academic journals.

Furthermore, in a case where an invention achieves the original useful purpose but the result of the invention would injure public health, an examiner is supposed to consider whether a method to eliminate the harmful effect of the present invention exists or whether the effect of the invention is advantageous or not.